THE ITERATION $x \mapsto a \bmod x$ HAS LENGTH $O_\varepsilon(a^{45/152+\varepsilon})$

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ABSTRACT. For integers $a > b > 0$ let $b_1 = b$, $b_{i+1} = a \bmod b_i$, and let $P(a,b)$ be the number of positive terms of this sequence. Shallit asked whether $P(a,b) = O((\log a)^2)$. The bound $P(a,b) \ll_\varepsilon a^{1/3+\varepsilon}$ of Erdős and Shallit was improved by Chase and Pandey to $a^{1/3-2/177+\varepsilon}$. We prove $P(a,b) \ll_\varepsilon a^{45/152+\varepsilon}$, where $45/152 = 0.29605\ldots$, by an elementary argument. We first prove $P(a,b) \ll_\varepsilon a^{3/10+\varepsilon}$. The new ingredient there is an exact identity for three consecutive steps of the iteration. It shows that a certain integer combination $D = dv - eu$ of the drops and quotient increments is always positive and is small when the drops are small. A second identity, for six consecutive steps, produces a further positive integer L. It gives the local bound $T \ll a^{1/2}A^{-1/3}$ for the number T of remainders in $[A, 2A)$, with no divisor-function loss. Then D and L are the third and fourth members of a hierarchy of positive integers, the Hankel determinants of an exact positive moment model of the orbit. To go below $3/10$ we use the arithmetic of L itself. At the critical scale L is a positive integer of bounded size. Seven such integers at two consecutive states form a pattern, and the states with a given pattern lie on a quadric whose quadratic part is nondegenerate, with at most four states per point. Counting its points by binary slices and norm equations gives the exponent $11/37$. Next the seven integers are Plücker coordinates of one matrix, and counting vectors of minors gives $8/27$. Finally the gap vector lies in a rank-three lattice determined by the minor vector. Weighting each minor vector by the shortest vector of its lattice, through an elementary reduced basis, gives $45/152$. Along the way we show $P(a,b) \ll a^{3/10}\sqrt{\log\log 3a}$, assuming results of Nair–Tenenbaum and de la Bretèche–Tenenbaum on the Erdős–Hooley Δ-function. We also show $P(a,b) \ll a^{1/4}\sqrt{R}$, where $R-1$ is the number of rises in the sequence of quotient increments. These results are formalised in Lean 4 with Mathlib, without proof holes. The $\sqrt{\log\log 3a}$ bound is formalised conditionally on the two cited theorems, which are stated in Lean as explicit hypotheses.

1. INTRODUCTION

Let $a > b > 0$ be integers. Put $b_1 = b$ and $b_{i+1} = a \bmod b_i$ while $b_i > 0$. The sequence is strictly decreasing until it reaches 0; if $b_{n+1} = 0$ we write $P(a,b) = n$. For example $P(35,22) = 7$, from $22, 13, 9, 8, 3, 2, 1, 0$. The number $P(a,b)$ is the length of the Pierce expansion of b/a [11, 12], since

$$\frac{b}{a} = \frac{1}{q_1} - \frac{1}{q_1 q_2} + \frac{1}{q_1 q_2 q_3} - \cdots, \qquad q_i = \lfloor a/b_i \rfloor.$$

Shallit [10] asked whether

$$P(a,b) = O\big((\log a)^2\big). \tag{1}$$

Shallit [12] proved $P(a,b) \ll a^{1/2}$ by "Archimedean" arguments. Erdős and Shallit [5] combined these with a divisor argument to get $P(a,b) \ll_\varepsilon a^{1/3+\varepsilon}$, and showed that $\max_b P(a,b)/\log a$ does not tend to 0. Chase and Pandey [4] went past the exponent $1/3$, proving $P(a,b) \ll_\varepsilon a^{1/3-2/177+\varepsilon}$ with exponential sum estimates (Bourgain's exponent pair), and showed that $\max_b P(a,b) \gg \log a/\log\log a$ for all large a. Baraskar and Vukusic [1] study the related sets of iterated remainders and bound their cardinalities at each fixed iteration depth. Shallit's talk pages [15] report a bound $n^{3/10}$, described as "LLM, unpublished so far".

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We have not seen that argument. The result below has the same exponent up to ε, and we make no claim about how the two arguments are related.

Theorem 1.1. *For every $\varepsilon > 0$ there is a constant C_ε such that*

$$P(a,b) \le C_\varepsilon\, a^{3/10+\varepsilon} \qquad \text{for all integers } 0 < b < a.$$

The proof is elementary. It uses only the division equations $a = q_i b_i + b_{i+1}$ and the divisor bound $\tau(n) \ll_\delta n^\delta$. As usual, the remainders are split into dyadic windows $[A, 2A)$, and we bound the number T of remainders in one window. There are two bounds:

$$T(T-1) \le 4A\,\Delta, \tag{2}$$

$$T^6 A^2 \le 995328\, a^3 \Delta \qquad (T \ge 12), \tag{3}$$

where Δ is the largest value of the divisor function on $[1, 32a^7]$. Bound (2) is the Erdős–Shallit argument. Bound (3) is new. Multiplying (3) by the square of (2) eliminates A and gives $T^{10} \ll a^3\Delta^3$, which is the exponent $3/10$.

The new bound comes from three consecutive steps $x = b_i$, $y = b_{i+1}$, $z = b_{i+2}$, $w = b_{i+3}$. Let d, e, f be the drops, let u, v be the increments of the quotients, and put $D = dv - eu$. Then

$$D\,z = df + ude + (u-1)e^2. \tag{4}$$

So D is a *positive* integer, even though dv and eu largely cancel, and it is small when the drops are small. Also $(q, x) = (\lfloor a/b_i \rfloor, b_i)$ lies on the curve $xM(q) = aN(q) + D$, where M and N are explicit polynomials depending on (u, v) (Lemma 3.3). A resultant-type identity shows that each such curve has $a^{o(1)}$ admissible points (Lemma 4.1). The number of triples (u, v, D) is $\ll (B/T)^2 \cdot a/T^3$, where $B = a/A$, which gives (3).

A second positive integer. Six consecutive terms carry three overlapping curvatures D, E, F and four quotient increments u, v, w, z. In Section 7 we show that the integer

$$L = \frac{S(v,w,z)\,D - S(u,v,w)\,E}{v}, \qquad S(u,v,w) = u^2 v + (u-1)v^2 + uw,$$

satisfies the exact identity $qwL = J(u,v,w,z)\,E + S(u,v,w)\,F$, where J is a polynomial with positive values (Lemma 7.1). Hence $L \ge 1$, and $q \asymp a/A$ is bounded by a polynomial in the small parameters u, v, w, z, E, F. A single window of six terms with small total drop then gives

$$T^6 A^2 \le 96 \cdot 10^6\, a^3 \qquad (T \ge 10), \tag{5}$$

with no divisor function at all (Proposition 7.3).

With (5) in hand, the only remaining loss is in the drop bound (2). That loss comes from the number of divisors of $a+h$ in $[A, 2A)$ as h ranges over a short interval. Section 8 shows that any bound $\ll y\,G(x)$ for such counts over intervals $(x, x+y]$ with $y \ge x^{1/4}$, uniform in A and with $G(x) \ll x^{1/40}$, gives $P(a,b) \ll a^{3/10} G(a)^{1/5}$ (Theorem 8.4). The counts are majorised by the Erdős–Hooley function. Combining the short-sum theorem of Nair and Tenenbaum [8] with the recent mean-value bound of de la Bretèche and Tenenbaum [2] gives $G(x) = (\log\log x)^{5/2}$, and hence the following.

Theorem 1.2. *Assume the theorem of Nair–Tenenbaum stated as* (9) *and the theorem of de la Bretèche–Tenenbaum stated as* (10) *below. Then*

$$P(a,b) \ll a^{3/10}\sqrt{\log\log 3a} \qquad \text{uniformly for integers } 0 < b < a.$$

Both hypotheses are published theorems. They are hypotheses only in the sense of the formalisation: we have not formalised their proofs. Any other short-interval divisor bound can be used in their place. For example, Shiu's theorem [14] gives $\sum_{x<n\le x+y}\tau(n) \ll y\log x$ for $y \ge x^{1/4}$, and hence $P(a,b) \ll a^{3/10}(\log a)^{1/5}$; we have not formalised that either.

Structured trajectories and the Hankel hierarchy. Write $k_i = q_{i+1} - q_i$ for the quotient increments, and let $R-1$ be the number of indices with $k_i < k_{i+1}$.

Theorem 1.3. *$P(a,b) \ll a^{1/4}\sqrt{R}$ uniformly for $0 < b < a$. In particular, if the quotient increments never rise, then $P(a,b) \ll a^{1/4}$.*

The proof (Section 9) combines three facts. First, $D \ge 1$ says that d_i/k_i is strictly decreasing, so the drops strictly decrease along every stretch without a rise. Second, the six-step integer L gives an exact identity for six quotients in arithmetic progression, which forces their common difference to be large (Lemma 9.1). Third, a counting lemma for sequences in which no value occurs too often. So an orbit of length about $a^{3/10}$ must contain $\gg a^{1/10}$ rises.

Finally, both D and L fit into a single structure (Section 10). The orbit carries an exact finite model by positive measures ν_i with $\int d\nu_i = b_i/a$. The Hankel determinants $I_r(i) = a\det\bigl(\int x^{j+l}\,d\nu_i\bigr)_{j,l<r}$ are positive integers, and I_1, I_2, I_3, I_4 are b_i, $b_i - b_{i+1}$, D and L (Theorem 10.2).

Below 3/10. We go past the exponent 3/10 in four steps.

Theorem 1.4. *For every $\varepsilon > 0$ there is a constant C_ε such that*

$$P(a,b) \le C_\varepsilon\, a^{23/77+\varepsilon} \qquad \text{for all integers } 0 < b < a.$$

Here $23/77 = 0.298701\ldots$, a saving of $1/770$ over $3/10$. The proof (Section 11) uses the fourth Hankel invariant $I_4 = L$ as an integer, not only as a positive number. At the critical configuration L has bounded size. The labels $L(i), L(i+1), L(i+2)$ and four cofactors of an explicit 3×4 integer matrix form a *pattern* of seven integers. Two consecutive states give two linear equations and one quadratic equation in five consecutive quotient increments, and the quadratic has a positive leading coefficient. An exact identity $I_4(i)\,\pi_i(-q_i) = a\det(\mu_i(x^{j+l}))_{j,l<4}$, with π_i the degree-four orthogonal polynomial of the moment model, makes q_i a root of a quartic that is fixed by the pattern and the increments. So each pattern accounts for $O(H^2)$ states, where H bounds the increments. The number of patterns is controlled by Hadamard's inequality applied to the moment model.

Counting the solutions of the quadratic equation more carefully gives a further improvement.

Theorem 1.5. *For every $\varepsilon > 0$ there is a constant C_ε such that*

$$P(a,b) \le C_\varepsilon\, a^{11/37+\varepsilon} \qquad \text{for all integers } 0 < b < a.$$

Here $11/37 = 0.297297\ldots$, a saving of $1/370$ over $3/10$. The extra input (Section 12) is that the quadratic part is nondegenerate with an indefinite binary restriction, by exact sign identities for the cofactors. So each slice is a binary norm equation, which has $a^{o(1)}$ solutions of size $a^{O(1)}$, uniformly in its coefficients, when its right-hand side is nonzero. There are at most two exceptional slices, which are counted separately.

Theorem 1.6. *For every $\varepsilon > 0$ there is a constant C_ε such that*

$$P(a,b) \le C_\varepsilon\, a^{8/27+\varepsilon} \qquad \text{for all integers } 0 < b < a.$$

Here $8/27 = 0.296296\ldots$. The seven pattern integers are signed coordinates of the vector of 3×3 minors of a 3×5 integer matrix (Section 13). All ten minors are bounded by X, and such minor vectors number $X^{5+o(1)}$ rather than X^7.

Theorem 1.7. *For every $\varepsilon > 0$ there is a constant C_ε such that*

$$P(a,b) \le C_\varepsilon\, a^{45/152+\varepsilon} \qquad \text{for all integers } 0 < b < a.$$

Here $45/152 = 0.296052\ldots$. The pivot gaps of a state lie in a lattice $\Lambda_p \subseteq \mathbb{Z}^3$ of index at most X determined by its minor vector p (Section 14). If the shortest vector of Λ_p has length λ, a reduced basis cuts the ternary quadratic into $O(H/\lambda + 1)$ slices, and only $X^{4+o(1)}t^3$ minor vectors have $\lambda \le t$. This gives $T \ll a^{o(1)}(HX^{14/3} + X^5)$ in place of $T \ll a^{o(1)}HX^5$.

The idea of encoding states by the coefficients of the degree-four orthogonal polynomial is due to an AI-assisted exploration (see the acknowledgements). It first gave a bound $a^{3/10-\delta}$ for a small $\delta > 0$, through the resultant identity $\operatorname{Res}(I_4\pi_4, I_5\pi_5) = aI_5^8$ for the orthogonal polynomials of degrees four and five, together with the dimension-growth bounds of Cluckers, Dèbes, Hendel, Nguyen and Vermeulen [3]. The two-state count of Section 11 replaces both by elementary algebra. The refinement to $11/37$, with the sign identities and the slice count, was also found in that exploration, as was the observation that the seven integers form a Plücker vector, and the proposal to weight each minor vector by the shortest vector of its lattice. The proof of the last step given here replaces the geometry of numbers by an elementary reduction argument.

Formal verification. Every step of the proofs of Theorems 1.1, 1.4, 1.5, 1.6 and 1.7 has been formalised in Lean 4 [6] using Mathlib [7]. So have Proposition 7.3, Theorem 8.4, the deduction of Theorem 1.2 from (9) and (10), Theorem 1.3 and Theorem 10.2. Section 15 describes the formalisation. The constants in this paper are the constants in the Lean proof. The source, independent submission statements and build instructions are available at `https://github.com/jasondavies/pierce-length-bound`.

What is not proved. These results are far from (1), and the method cannot reach it; see Section 16.

2. Preliminaries

2.1. **Notation.** Throughout, $a > b > 0$ are integers, $b_1 = b$, $b_{i+1} = a \bmod b_i$, and $P = P(a,b)$, so that $a > b_1 > b_2 > \cdots > b_P > 0 = b_{P+1}$. Write $q_i = \lfloor a/b_i \rfloor$, so

$$a = q_i b_i + b_{i+1}, \qquad 0 \le b_{i+1} < b_i \qquad (1 \le i \le P).$$

Let $\tau(n)$ be the number of positive divisors of n, and let

$$\Delta = \Delta(a) = \max_{1 \le n \le 32a^7} \tau(n).$$

Lemma 2.1 (Quotients increase)**.** *If $b_{i+1} > 0$ then $q_{i+1} \ge q_i + 1$.*

Proof. Since $b_{i+1} < b_i$ we have $(q_i+1)b_{i+1} \le q_i b_i + b_{i+1} = a$, so $q_{i+1} = \lfloor a/b_{i+1} \rfloor \ge q_i + 1$. □

Lemma 2.2 (Divisor bound)**.** *For every integer $k \ge 1$ there is C_k with $\tau(n)^k \le C_k\, n$ for all $n \ge 1$.*

Proof. This is standard: $\tau(n)^k/n = \prod_{p^\alpha \| n} (\alpha+1)^k/p^\alpha$. Each factor with $p \ge 2^k$ is at most $(\alpha+1)^k/2^{k\alpha} \le 1$. Each of the fewer than 2^k factors with $p < 2^k$ is at most $\sup_\alpha (\alpha+1)^k/2^\alpha < \infty$. □

2.2. **Window runs.**

Definition 2.3. Let $A \geq 1$ be an integer. A *window run* of length T is a sequence $x_0, x_1, \ldots, x_{T-1}$ of consecutive terms of the sequence (b_i), so $x_{j+1} = a \bmod x_j$, all lying in $[A, 2A)$.

In a window run, $x_0 > x_1 > \cdots > x_{T-1}$ and $x_j \leq a$.

3. The curvature of three consecutive steps

Fix four consecutive terms

$$x = b_i, \quad y = b_{i+1} = a \bmod x, \quad z = b_{i+2} = a \bmod y, \quad w = b_{i+3} = a \bmod z,$$

with $x, y, z > 0$ (and $w \geq 0$). Write

$$q = \left\lfloor \frac{a}{x} \right\rfloor, \quad u = \left\lfloor \frac{a}{y} \right\rfloor - q, \quad v = \left\lfloor \frac{a}{z} \right\rfloor - \left\lfloor \frac{a}{y} \right\rfloor, \qquad d = x - y, \quad e = y - z, \quad f = z - w.$$

By Lemma 2.1, $u, v \geq 1$, and clearly $d, e, f \geq 1$. The division equations read

$$xq + y = a, \qquad y(q+u) + z = a, \qquad z(q+u+v) + w = a. \tag{6}$$

Definition 3.1. The *curvature* of the block is $D = dv - eu$.

Lemma 3.2 (Curvature identity)**.**

$$D\,z = df + ude + (u-1)e^2.$$

In particular $D \geq 1$.

Proof. Subtracting the first two equations of (6) gives $uy = qd + e$, hence $uz = uy - ue = qd + (1-u)e$. Subtracting the last two gives $vz = (q+u)e + f$. Therefore

$$(dv - eu)z = d\big((q+u)e + f\big) - e\big(qd + (1-u)e\big) = df + ude + (u-1)e^2.$$

The right-hand side is at least $df \geq 1$, and $z > 0$, so $D > 0$. □

Equivalently, the identity is the linear combination

$$Dz - \big(df + ude + (u-1)e^2\big) = d\,E_3 + (a - qx - x - y + z)\,E_2 + (-a + (q+u)y + y)\,E_1$$

of the three division equations E_1, E_2, E_3 of (6) (each written as left $-$ right $= 0$). This is how it is checked in Lean.

Lemma 3.3 (The fiber equation)**.** *Put* $M(q) = uq^2 + (u^2+u+v)q + v$ *and* $N(q) = uq + u^2 + v$. *Then*

$$x\,M(q) = a\,N(q) + D.$$

Proof. From (6), $d = (q+1)x - a$ and $e = y - z = (q+u+1)y - a = a(q+u) - q(q+u+1)x$. Substituting these into $D = dv - eu$ gives $D = x\big(uq(q+u+1) + v(q+1)\big) - a\big(u(q+u) + v\big)$, which is the claim. □

Lemma 3.4 (Size of the parameters)**.** *If* $x \leq a$ *then*

$$u\,yx \leq a(d+e), \qquad v\,zy \leq a(e+f), \qquad D\,z \leq (u+1)(d+e+f)^2.$$

Proof. From $uy = qd + e$ we get $uyx = (qx)d + ex \leq ad + ae$, since $qx \leq a$ and $x \leq a$. Similarly $vz = (q+u)e + f$ with $(q+u)y \leq a$ and $y \leq a$ gives the second bound. For the third, by Lemma 3.2, $Dz = df + ude + (u-1)e^2 \leq df + u\,e(d+e) \leq (1+u)(d+e+f)^2$. □

4. Few blocks share a triple (u, v, D)

Lemma 4.1 (Fiber count)**.** *Let $a \geq 1$, $1 \leq u, v \leq a$ and $1 \leq D \leq 2a$. Let $M(q) = uq^2 + pq + v$ with $p = u^2 + u + v$, and let $L(q) = cq + \ell$ with $c = au$ and $\ell = a(u^2 + v) + D$. Then the set*

$$Q = \{q \geq 0 : \ xM(q) = L(q) \text{ for some integer } x \geq 1\}$$

has at most $\Delta = \max_{n \leq 32a^7} \tau(n)$ elements.

Proof. Put $U(q) = ucq + cp - u\ell$ and $R = u\ell^2 - cp\ell + c^2 v$. Expanding shows

$$c^2 M(q) - U(q)L(q) = R \tag{7}$$

identically in q. If $q \in Q$ with witness x, substituting $L(q) = xM(q)$ into (7) gives

$$R = M(q)\big(c^2 - U(q)x\big). \tag{8}$$

We bound the sizes first. From $u, v \leq a$ and $D \leq 2a$ we get $\ell \leq 4a^3$, $p \leq 3a^2$ and $c \leq a^2$, so

$$|R| \leq u\ell^2 + cp\ell + c^2 v \leq 16a^7 + 12a^7 + a^5 \leq 32a^7, \qquad c^2 \leq a^4 \leq 32a^7.$$

Case $R \neq 0$. By (8), $M(q)$ is a positive divisor of $|R|$. Since $u, p \geq 1$, the map $q \mapsto M(q)$ is strictly increasing on $q \geq 0$, hence injective. Therefore $|Q| \leq \tau(|R|) \leq \Delta$.

Case $R = 0$. Since $M(q) > 0$, (8) gives $U(q)x = c^2$. Here $c^2 > 0$ and $x \geq 1$, so $U(q)$ is a positive divisor of c^2. Since U is linear with positive leading coefficient uc, it is injective. Therefore $|Q| \leq \tau(c^2) \leq \Delta$. □

The case $R = 0$ cannot be dropped: it is the case in which M and L might share a factor.

5. Counting visits to a window

In this section $x_0, \ldots, x_{T-1}$ is a window run in $[A, 2A)$, and $\Delta = \max_{n \leq 32a^7} \tau(n)$.

Proposition 5.1 (Drop bound; after Erdős–Shallit)**.** *If $T \geq 2$ then $T(T-1) \leq 4A\Delta$.*

Proof. Let $h_j = x_j - x_{j+1} \geq 1$ for $0 \leq j \leq T-2$. By telescoping, $\sum_j h_j = x_0 - x_{T-1} \leq (2A-1) - A = A - 1$. Call j *good* if $h_j(T-1) \leq 2A$ and *bad* otherwise. Each bad j has $h_j(T-1) \geq 2A+1$, so

$$|\operatorname{Bad}|\,(2A+1) \leq (T-1)\textstyle\sum_j h_j \leq (T-1)(A-1).$$

If $2|\operatorname{Bad}| \geq T-1$ this would give $(T-1)(2A+1) \leq 2(T-1)(A-1)$, which is false. Hence $2|\operatorname{Bad}| < T-1$, and since $|\operatorname{Good}| + |\operatorname{Bad}| = T-1$ we get $2|\operatorname{Good}| \geq T$.

For good j we have $1 \leq h_j \leq H := \lfloor 2A/(T-1) \rfloor$. Also $a + h_j = q_j x_j + x_{j+1} + x_j - x_{j+1} = (q_j + 1)x_j$, so $x_j \mid a + h_j$. Moreover $a + h_j \leq 2a \leq 32a^7$. Since $j \mapsto x_j$ is injective, at most $\tau(a+h) \leq \Delta$ good indices have $h_j = h$. Hence $|\operatorname{Good}| \leq H\Delta$, and

$$T(T-1) \leq 2|\operatorname{Good}|(T-1) \leq 2\Delta\, H(T-1) \leq 4A\Delta.$$ □

Proposition 5.2 (Curvature bound)**.** *If $T \geq 12$ then $T^6 A^2 \leq 995328\, a^3 \Delta$.*

Proof. For $0 \leq j \leq T-4$ let $s_j = x_j - x_{j+3}$ be the total drop of the block $x_j, x_{j+1}, x_{j+2}, x_{j+3}$. By telescoping,

$$\sum_{j=0}^{T-4} s_j = (x_0 + x_1 + x_2) - (x_{T-3} + x_{T-2} + x_{T-1}) \leq 3(2A-1) - 3A = 3A - 3.$$

Call j *good* if $s_jT \le 12A$ and *bad* otherwise. Then $|\,\mathrm{Bad}\,|(12A+1) \le T(3A-3)$. If $4|\,\mathrm{Bad}\,| \ge T$ this would give $T(12A+1) \le 4T(3A-3)$, which is false. So $4|\,\mathrm{Bad}\,| \le T-1$. Since $|\,\mathrm{Good}\,| = T-3-|\,\mathrm{Bad}\,|$ and $T \ge 12$,

$$2|\,\mathrm{Good}\,| \ge 2T-6-\tfrac{T-1}{2} \ge T.$$

Fix a good j. Apply Section 3 to $x = x_j$, so that $y, z, w = x_{j+1}, x_{j+2}, x_{j+3}$, all in $[A, 2A)$. Write $s = s_j = d+e+f$. Lemma 3.4 gives $uA^2 \le uyx \le a(d+e) \le as$. Multiplying by T and using $sT \le 12A$ gives $uA^2T \le 12aA$, that is,

$$u\,AT \le 12a, \qquad \text{and likewise} \qquad v\,AT \le 12a.$$

Also $DA \le Dz \le (u+1)s^2 \le 2us^2$. Multiplying by T^2 gives $DAT^2 \le 2u(sT)^2 \le 288uA^2$, so $DT^2 \le 288uA$, and then

$$D\,T^3 \le 288\,u\,AT \le 3456\,a.$$

Thus the triple (u, v, D) of a good block lies in $[1, U]^2 \times [1, D^*]$, where

$$U = \lfloor 12a/(AT) \rfloor \le a, \qquad D^* = \lfloor 3456a/T^3 \rfloor \le 2a.$$

The inequalities $U \le a$ and $D^* \le 2a$ hold because $AT \ge 12$ and $T^3 \ge 1728$.

Now fix such a triple and consider the good blocks that have it. By Lemma 3.3, each one gives a point $q = q_j = \lfloor a/x_j \rfloor$ of the set Q of Lemma 4.1. The map $j \mapsto q_j$ is injective on these blocks: q_j determines $x_j = L(q_j)/M(q_j)$, and x_j determines j. By Lemma 4.1, at most Δ good blocks share a triple. So $|\,\mathrm{Good}\,| \le U^2D^*\Delta$, and

$$T^6A^2 = T\,(AT)^2\,T^3 \le 2\Delta\,(U\,AT)^2\,(D^*T^3) \le 2\Delta(12a)^2(3456a) = 995328\,a^3\Delta. \qquad \square$$

Corollary 5.3 (Window bound)**.** *If $T \ge 12$ then $T^{10} \le 63700992\,a^3\Delta^3$.*

Proof. Using $T \le 2(T-1)$ and the two propositions,

$$T^{10}A^2 \le 4\,(T^6A^2)\,\big(T(T-1)\big)^2 \le 4 \cdot 995328\,a^3\Delta \cdot (4A\Delta)^2 = 63700992\,a^3\Delta^3A^2.$$

Divide by A^2. $\square$

Remark 5.4. In terms of $B = a/A$, Propositions 5.1 and 5.2 say $T \ll a^{1/2+o(1)}B^{-1/2}$ and $T \ll a^{1/6+o(1)}B^{1/3}$. The first is better when $B > a^{2/5}$, and both equal $a^{3/10+o(1)}$ at $B = a^{2/5}$. Corollary 5.3 takes the product of the two bounds, so no case split is needed.

6. Proof of Theorem 1.1

Let $0 < b < a$. For $0 \le k \le \log_2 a$ let T_k be the number of indices $i \le P$ with $2^k \le b_i < 2^{k+1}$. Every b_i with $i \le P$ lies in $[1, a)$, so $P = \sum_{0 \le k \le \log_2 a} T_k$. Since (b_i) is decreasing, the indices counted by T_k are consecutive. So when $T_k > 0$ they form a window run in $[A, 2A)$ with $A = 2^k$.

Let $\varepsilon > 0$. Choose an integer $m > 7/(5\varepsilon)$, and let C_{3m} be the constant of Lemma 2.2 with $k = 3m$. The maximum Δ is attained at some $n_0 \le 32a^7$, so $\Delta^{3m} \le C_{3m} \cdot 32a^7$. If $T_k \ge 12$, Corollary 5.3 gives

$$T_k^{10m} \le 63700992^m\,a^{3m}\,\Delta^{3m} \le K\,a^{3m+7}, \qquad K = 63700992^m \cdot C_{3m} \cdot 32,$$

and hence

$$T_k \le K^{1/(10m)}\,a^{3/10+7/(10m)} \le K^{1/(10m)}\,a^{3/10+\varepsilon/2}.$$

If $T_k \le 11$ this bound holds with 11 in place of $K^{1/(10m)}$. Finally $\lfloor \log_2 a \rfloor + 1 \le C'_\varepsilon a^{\varepsilon/2}$. Summing over k,

$$P(a,b) \le (\lfloor \log_2 a \rfloor + 1)\big(11 + K^{1/(10m)}\big)a^{3/10+\varepsilon/2} \le C_\varepsilon\,a^{3/10+\varepsilon}. \qquad \square$$

7. Six consecutive steps

Fix six consecutive terms $x_0 > x_1 > \cdots > x_5 \geq 0$ of the sequence (b_i) with $x_4 > 0$, and write their quotients as

$$\lfloor a/x_0 \rfloor = q, \quad \lfloor a/x_1 \rfloor = q+u, \quad \lfloor a/x_2 \rfloor = q+u+v,$$
$$\lfloor a/x_3 \rfloor = q+u+v+w, \quad \lfloor a/x_4 \rfloor = q+u+v+w+z.$$

(In this section w and z are quotient increments, not terms of the sequence.) By Lemma 2.1, $u, v, w, z \geq 1$. Let $f_k = x_k - x_{k+1} \geq 1$ be the drops, and let

$$D = f_0 v - f_1 u, \qquad E = f_1 w - f_2 v, \qquad F = f_2 z - f_3 w$$

be the curvatures of the blocks starting at x_0, x_1 and x_2. Write $\mathcal{E}_k$ for the division equation $x_k q_k + x_{k+1} - a = 0$, where $q_k = \lfloor a/x_k \rfloor$. Put

$$S(u,v,w) = u^2 v + (u-1)v^2 + uw,$$
$$J(u,v,w,z) = \big(u^2 + (u-1)v\big)z + w\big(u^3 + uv(2u-1) + (u-1)v^2\big) + w^2\big((u-1)(u+v)+1\big),$$
$$L = (vw + w^2 + z)\, D - \big(u^2 + (u-1)v\big)E - w\,(wf_0 - uf_2).$$

Lemma 7.1 (The six-step identity)**.** (1) $S(v,w,z)\, D - S(u,v,w)\, E = v\, L$.
(2) $q\, w\, L = J(u,v,w,z)\, E + S(u,v,w)\, F$.

Proof. Part (1) is a polynomial identity: since $S(v,w,z) \equiv -w^2$ and $S(u,v,w) \equiv uw \pmod v$, and $wD + uE = v(wf_0 - uf_2)$, the difference is divisible by v, and expanding gives L. Part (2) holds modulo the division equations. Explicitly, $qwL - JE - S(u,v,w)F$ equals

$$w(v^2 w + vw^2 + vz - w^2)\,\mathcal{E}_0 - w(u^2 w + 2uvw + uw^2 + uz + v^2 w + vw^2 - vw + vz - w^2)\,\mathcal{E}_1$$
$$+ w(u^2 v + u^2 w + uv^2 + 2uvw + uw^2 + uw + uz - v^2 - vw)\,\mathcal{E}_2 - w(u^2 v + uv^2 + uw - v^2)\,\mathcal{E}_3.$$

This is an identity of polynomials in a, q, the x_k and the increments, checked by expansion. □

Corollary 7.2. *Suppose $q \geq 1$ and $u, v, w, z \leq m$. Then $E, F \geq 1$, $L \geq 1$, and*

$$q \leq 9m^3 E + 3m^3 F.$$

Proof. Lemma 3.2, applied to the blocks starting at x_1 and x_2, gives $E, F \geq 1$. Clearly $J > 0$ and $S \geq 0$, so by Lemma 7.1(2) $qwL > 0$. Since L is an integer, $L \geq 1$ and $qw \leq JE + SF$. Bounding each monomial gives $J \leq 9m^3 w$ and $S(u,v,w) \leq 3m^3 w$. Dividing by w gives the claim. □

So the quotient q, which is about a/A, is controlled by the small parameters of a single block of six terms. No counting over many blocks is needed.

Proposition 7.3 (Six-step window bound)**.** *For a window run of length $T \geq 10$,*

$$T^6 A^2 \leq 96 \cdot 10^6\, a^3.$$

Proof. For $0 \leq j \leq T-6$ let $s_j = x_j - x_{j+5}$. By telescoping, $\sum_{j \leq T-6} s_j \leq 5(2A-1) - 5A = 5A-5$. So some j has $s_j(T-5) \leq 5A-5$, and since $T \leq 2(T-5)$, we get $s := s_j$ with $sT \leq 10A$. Apply Corollary 7.2 to $x_j, \ldots, x_{j+5}$, all of which lie in $[A, 2A)$, with $m = \max(u,v,w,z)$.

By Lemma 3.4 at x_j, x_{j+1} and x_{j+2}, each increment satisfies (increment) $\cdot\, A^2 \leq as$. Multiplying by T gives $mAT \leq 10a$. By Lemma 3.4 again, $EA \leq Ex_{j+3} \leq (v+1)s^2 \leq 2ms^2$, and likewise $FA \leq 2ms^2$. Hence $qA \leq 24m^4 s^2$. Also $a = qx_j + x_{j+1} < (q+1)x_j < 2A(q+1) \leq 4Aq$. Therefore

$$qA \cdot A^4 T^6 \leq 24\,(mAT)^4 (sT)^2 \leq 24 \cdot 10^6\, a^4 A^2,$$

so $qA^3 T^6 \leq 24 \cdot 10^6 a^4$, and $aA^2 T^6 \leq 4qA^3 T^6 \leq 96 \cdot 10^6\, a^4$. □

Remark 7.4. In terms of $B = a/A$, Proposition 7.3 reads $T \ll a^{1/6}B^{1/3}$. This is Proposition 5.2 without the factor $\Delta^{1/6}$. Combined with Proposition 5.1 as in Corollary 5.3, it gives $T^{10} \ll a^3\Delta^2$ and another proof of Theorem 1.1. The Lean proof of Theorem 1.1 uses Corollary 5.3.

8. Divisors over short intervals

For $A \geq 1$ let $\tau_A(n) = \#\{d \mid n : \ A \leq d < 2A\}$.

Proposition 8.1 (Drop count)**.** *For a window run of length $T \geq 2$ and any $Y \geq \lfloor 2A/(T-1) \rfloor$,*

$$T \leq 2 \sum_{a<n\leq a+Y} \tau_A(n).$$

Proof. This is the proof of Proposition 5.1, stopped before the divisor bound. The good indices j have $1 \leq h_j \leq \lfloor 2A/(T-1) \rfloor$, and $j \mapsto x_j$ maps those with $h_j = h$ injectively into the divisors of $a + h$ in $[A, 2A)$. □

Definition 8.2. Let $G\colon \mathbb{N} \to \mathbb{R}$. We say that $\mathrm{SD}(G)$ holds if there are C, x_0 such that for all integers $x \geq x_0$, $x^{1/4} \leq y \leq x$ and $A \geq 1$,

$$\sum_{x<n\leq x+y} \tau_A(n) \leq C\, y\, G(x).$$

Lemma 8.3 (Geometric sums)**.** *Let $\alpha, \beta, s, t > 0$ and $K \geq 0$. Then*

$$\sum_{k<K} \min\bigl(\alpha 2^{-sk},\ \beta 2^{tk}\bigr) \leq \Bigl(\frac{1}{1-2^{-s}} + \frac{2^t}{2^t-1}\Bigr)\alpha^{t/(s+t)}\beta^{s/(s+t)}.$$

Proof. Let $M = \alpha^{t/(s+t)}\beta^{s/(s+t)}$. Then $(\alpha 2^{-sk})^t(\beta 2^{tk})^s = M^{s+t}$, so for each k either $\beta 2^{tk} \leq M$ or $\alpha 2^{-sk} < M$. The terms $\beta 2^{tk} \leq M$ grow with ratio 2^t, so they sum to at most $M2^t/(2^t-1)$. The terms $\alpha 2^{-sk} < M$ decay with ratio 2^{-s}, so they sum to at most $M/(1-2^{-s})$. □

Theorem 8.4. *Suppose $\mathrm{SD}(G)$ holds, $G \geq 1$, and $G(x) \ll x^{1/40}$. Then $P(a,b) \ll a^{3/10}\, G(a)^{1/5}$ uniformly for $0 < b < a$.*

Proof. Small a are handled by $P(a,b) \leq b < a$. Fix a dyadic window $[A, 2A)$, $A = 2^k$, with $T \geq 20$ visits. Proposition 7.3 gives $T \leq \alpha A^{-1/3}$, where $\alpha = (96 \cdot 10^6)^{1/6}a^{1/2}$. Take $Y = \max(\lfloor 2A/(T-1) \rfloor, \lceil a^{1/4} \rceil) \leq a$ in Proposition 8.1. With $H = \lfloor 2A/(T-1) \rfloor$, $\mathrm{SD}(G)$ gives $T \leq 2CG(a)(H + a^{1/4} + 1)$. Multiplying by $T-1$, using $H(T-1) \leq 2A$, and solving the quadratic gives $T \leq 1 + 2CG(a)(a^{1/4}+1) + 2\sqrt{CG(a)A}$, hence, as $T \geq 20$, $T \leq \gamma + \beta A^{1/2}$, where $\beta = 4\sqrt{CG(a)}$ and $\gamma = 4CG(a)(a^{1/4}+1)$. So every window satisfies

$$T_k \leq 19 + \gamma + \min\bigl(\alpha\, 2^{-k/3},\ \beta\, 2^{k/2}\bigr).$$

Lemma 8.3 with $s = 1/3$ and $t = 1/2$, where $1/(1-2^{-1/3}) + \sqrt{2}/(\sqrt{2}-1) < 9$, gives

$$\sum_k \min\bigl(\alpha\, 2^{-k/3},\ \beta\, 2^{k/2}\bigr) \leq 9\, \alpha^{3/5}\beta^{2/5} \ll a^{3/10}G(a)^{1/5},$$

with no factor $\log a$. The remaining terms contribute $(\log_2 a + 1)(19 + \gamma) \ll a^{1/100} \cdot a^{1/4}G(a)$. This is $\ll a^{3/10}G(a)^{1/5}$, because $G(a)^{4/5} \ll a^{1/50}$. □

The Erdős–Hooley function. Let $\Delta_{\mathrm{H}}(n) = \max_{u\in\mathbb{R}} \#\{d \mid n : \ e^u < d \le e^{u+1}\}$. Taking $e^{u+1} = 2A$, and using $e > 2$, gives $\tau_A(n) \le \Delta_{\mathrm{H}}(n)$. We use two published theorems.

(NT) Nair and Tenenbaum [8] prove a general upper bound for short sums of functions g satisfying $g(mn) \le \min(\lambda_1^{\Omega(m)}, \lambda_2 m^{\varepsilon})g(n)$ for $(m,n) = 1$. Since $\Delta_{\mathrm{H}}(mn) \le \tau(m)\Delta_{\mathrm{H}}(n)$ for $(m,n) = 1$, the function Δ_{H} satisfies this with $\lambda_1 = 2$ for every $\varepsilon > 0$. Their bound, for the polynomial $Q(X) = X$, gives for every fixed $\sigma > 0$, x large and $x^\sigma \le y \le x$

$$\sum_{x<n\le x+y} \Delta_{\mathrm{H}}(n) \ll_\sigma \frac{y}{\log x} \sum_{n\le x} \frac{\Delta_{\mathrm{H}}(n)}{n}. \tag{9}$$

We use it with $\sigma = 1/4$. In the parameters of their Theorem 1, take $g = 1$, $\varepsilon = 1/16$ and $\delta = 1$: then $4g^2\varepsilon = 1/4$ and the required class is $\mathcal{M}_1(2, B, 1/48)$ for a suitable B. The same consequence is displayed immediately after equation (2) of their paper.

(BT) de la Bretèche and Tenenbaum [2, Theorem 1.1] prove that for large x

$$\sum_{n\le x} \Delta_{\mathrm{H}}(n) \ll x\,(\log\log x)^{5/2}. \tag{10}$$

Lemma 8.5. *Assume* (10)*, and let* $\ell(x) = \max(1, \log\log x)$*. Then* $\sum_{n\le x} \Delta_{\mathrm{H}}(n)/n \ll \log x \cdot \ell(x)^{5/2}$ *for large* x*.*

Proof. Split n into dyadic blocks $2^j \le n < 2^{j+1} \le 2x$. The terms with n below the threshold of (10) contribute $O(1)$, since $\Delta_{\mathrm{H}}(n) \le n$. Each remaining block contributes at most $2^{-j}\sum_{n\le 2^{j+1}} \Delta_{\mathrm{H}}(n) \ll \ell(2x)^{5/2}$, and $\ell(2x) \le 2\ell(x)$. There are at most $\log_2 x + 1 \le 3\log x$ blocks. □

Proof of Theorem 1.2. By (9) and Lemma 8.5, $\sum_{x<n\le x+y} \tau_A(n) \le \sum_{x<n\le x+y} \Delta_{\mathrm{H}}(n) \ll y\,\ell(x)^{5/2}$ for $x^{1/4} \le y \le x$. So SD(G) holds with $G = \ell^{5/2}$, which satisfies $G \ge 1$ and $G(x) \ll x^{1/40}$. Theorem 8.4 gives $P(a,b) \ll a^{3/10}\ell(a)^{1/2} \ll a^{3/10}\sqrt{\log\log 3a}$. □

9. Rises in the quotient increments

Throughout this section $k_i = q_{i+1} - q_i \ge 1$, $d_i = b_i - b_{i+1} \ge 1$, and a *rise* is an index with $k_i < k_{i+1}$.

Lemma 9.1 (Six quotients in arithmetic progression)**.** *Let* $q, h \ge 1$ *and* $x_0 > x_1 > \cdots > x_6 \ge 0$ *satisfy* $x_j(q + jh) + x_{j+1} = a$ *for* $0 \le j \le 5$*. Then* $q^4 \le 324\,h^3 a$*.*

Proof. Let $f_j = x_j - x_{j+1}$, and let $E = (f_1 - f_2)h$ and $F = (f_2 - f_3)h$ be the curvatures of the blocks starting at x_1 and x_2 (all their increments equal h). With $U = f_0 - 2f_1 + f_2$, the division equations $\mathcal{E}_j\colon x_j(q + jh) + x_{j+1} - a = 0$ give the exact identity

$$qhU = (3h-1)E + F, \qquad \text{namely} \qquad qhU - (3h-1)E - F = h(\mathcal{E}_0 - 3\mathcal{E}_1 + 3\mathcal{E}_2 - \mathcal{E}_3).$$

This is the case $u = v = w = z = h$ of Lemma 7.1, where $L = 2h^3U$. By Lemma 3.2, $E, F \ge 1$, so the integer U is at least 1.

Now bound E and F. Subtracting consecutive division equations gives $q_j f_j \le h x_{j+1}$, where $q_j = q + jh$. Since $x_{j+1}q_{j+1} \le a$, this yields $q^2 f_j \le ha$ for $1 \le j \le 4$. By Lemma 3.4, $Ex_3 \le (h+1)(f_1+f_2+f_3)^2 \le 9h^2(h+1)a^2/q^4$. Also $a < x_3(q+3h+1)$, so $Eq^4 \le 9h^2(h+1)(q+3h+1)a$, and likewise $Fq^4 \le 9h^2(h+1)(q+4h+1)a$. Hence

$$q^5 h \le q^4\big((3h-1)E + F\big) \le 27h^3(h+1)(q+4h+1)a.$$

If $q \ge h$ this gives $q^4 \le 324h^3a$. If $q < h$ then $q^4 < h^4 \le h^3 a$, since $h \le q_1 \le a$. □

Remark 9.2. An earlier version of this argument proved Lemma 9.1 for seven quotients by an analytic argument: it represents a periodic continuation of the orbit by integrals of positive functions, and compares finite differences. The identity above gives an algebraic proof with one fewer quotient.

Lemma 9.3. *Whenever $b_{i+2} > 0$, $d_{i+1}k_i < d_i k_{i+1}$. In particular, if $k_{i+1} \le k_i$ then $d_{i+1} < d_i$.*

Proof. This is $D = d_i k_{i+1} - d_{i+1}k_i \ge 1$ (Lemma 3.2). If $k_{i+1} \le k_i$ and $d_{i+1} \ge d_i$, then $d_i k_{i+1} \le d_{i+1}k_i$, which is a contradiction. □

Lemma 9.4. *Let $c_j \ge 1$ be integers, $j \in J$, such that no value occurs more than M times. Then $|J|^2 \le 2M \sum_{j\in J} c_j$.*

Proof. Induct on $|J|$. Let c_{j_0} be the largest value. The values lie in $[1, c_{j_0}]$, so $|J| \le M c_{j_0}$. Removing j_0 and using $|J|^2 = (|J|-1)^2 + 2|J| - 1$ completes the induction. □

Proposition 9.5. *Let $x_0, \dots, x_{T-1}$ be a window run in $[A, 2A)$ with $T \ge 2$, and let ρ be the number of rises among its $T-1$ increments. Then:*

(1) $(T-1)^2 \le 2(\rho+1)(A-1)$;
(2) $T - 1 \le \lfloor a/A \rfloor$;
(3) *if $6A \le a$, then $T - 1 = \ell + s$ with $\ell^2 \le 8(\rho+1)\lfloor a/A \rfloor$ and $s^3 \le 26244\, A$.*

Proof. Split the increments into *runs*: maximal stretches containing no rise. There are at most $\rho + 1$ runs.

(1) By Lemma 9.3, the drops strictly decrease along a run, so each value of d_j occurs at most once per run, hence at most $\rho + 1$ times. Lemma 9.4 and $\sum_j d_j = x_0 - x_{T-1} \le A - 1$ give the claim.

(2) The increments are ≥ 1, and they sum to $q_{T-1} - q_0 \le \lfloor a/A \rfloor$.

(3) Call an increment h *low* if $26244h^3A^4 < a^3$. Five consecutive equal increments h inside the run give six quotients in arithmetic progression. Lemma 9.1 then gives $q^4 \le 324h^3a$. On the other hand, $a < 3Aq$, which follows from $a = qx_0 + x_1$, $x_1 < x_0 < 2A$ and $6A \le a$. Together these give $a^3 \le 26244h^3A^4$, so h is not low. Within a run the increments are nonincreasing, so equal values are consecutive. Hence each low value occurs at most 4 times per run, and at most $4(\rho+1)$ times in all. Lemma 9.4 bounds the number ℓ of low increments by $\ell^2 \le 8(\rho+1)\sum_j k_j \le 8(\rho+1)\lfloor a/A \rfloor$. Let s be the number of high increments, with smallest value h_*. Then $sh_* \le \lfloor a/A \rfloor$ and $a^3 \le 26244h_*^3A^4$, so $s^3a^3 \le 26244A^4(a/A)^3 \le 26244Aa^3$. □

Proof of Theorem 1.3. The rises inside a window are rises of the orbit, so $\rho + 1 \le R$. Combining the three parts of Proposition 9.5, as in the proof of Theorem 8.4, gives for the window $[2^k, 2^{k+1})$ (when $6A > a$, part (2) gives $T - 1 \le a/A < 6 \le 30A^{1/3}$)

$$T_k \le 1 + \sqrt{R}\,\min\bigl(\sqrt{8a}\,2^{-k/2},\ \sqrt{2}\,2^{k/2}\bigr) + \min\bigl(a\,2^{-k},\ 30\cdot 2^{k/3}\bigr).$$

Lemma 8.3 gives $\sum_k \min(\sqrt{8a}\,2^{-k/2}, \sqrt{2}\,2^{k/2}) \ll (\sqrt{8a}\sqrt{2})^{1/2} = 2a^{1/4}$. It also gives $\sum_k \min(a2^{-k}, 30\cdot 2^{k/3}) \ll a^{1/4}$. Since $\log_2 a + 1 \ll a^{1/4}$, summing over k gives the theorem. □

Remark 9.6. By Theorem 1.3, an orbit with $P(a,b) \asymp a^{3/10}$ must have $R \gg a^{1/10}$. Proposition 9.5(3) says more: inside the critical window $A \asymp a^{3/5}$, a run of $T \asymp a^{3/10}$ remainders must contain $\gg a^{1/5}$ rises.

10. The Hankel hierarchy

Let $b_0 > b_1 > \cdots > b_{N-1} > 0$ be the positive part of an orbit, with quotients q_i and increments $k_i = q_{i+1} - q_i$. From here on the orbit is indexed from 0, so $b_0 = b$, $N = P(a, b)$ and $b_N = 0$; the symbols p_i, w_i below are unrelated to p and w of the earlier sections. Put $p_i(j) = q_{i+j} - q_i$ for $j < N - i$. Define weights $w_i(j)$ backwards by $w_{N-1} = \delta_0$ and

$$w_i(0) = 1 - \sum_j \frac{w_{i+1}(j)}{p_i(j+1)}, \qquad w_i(j+1) = \frac{w_{i+1}(j)}{p_i(j+1)}.$$

Let μ_i be the measure with mass $w_i(j)$ at $p_i(j)$, and let $\nu_i = \mu_i/(q_i + x)$. Write $h_i(s) = \int x^s \, d\nu_i = \sum_j w_i(j) p_i(j)^s / q_{i+j}$.

Lemma 10.1. (1) *The weights are positive and sum to* 1.

(2) $h_i(0) = b_i/a$, *that is,* $b_i/a = \mathbb{E}\big[1/(q_i + X_i)\big]$ *for* $X_i \sim \mu_i$; *and* $h_{N-1}(s) = 0$ *for* $s \geq 1$.

(3) $h_i(s+1) = \sum_{t \leq s} \binom{s}{t} k_i^{s-t} h_{i+1}(t)$ *for* $i + 1 < N$.

(4) $h_i(s) = (-q_i)^s b_i/a + z$ *with* $z \in \mathbb{Z}$.

Proof. (1) Since $p_i(j+1) \geq 1$, the sum defining $w_i(0)$ is at most 1. It is strictly less: if $N - i - 1 \geq 2$, the term $j = 1$ has $p_i(2) \geq 2$; if $i = N - 2$, the last gap satisfies $q_{N-1} \geq q_{N-2} + 2$, because $b_{N-1} \mid a$ gives $q_{N-1} - q_{N-2} - 1 = q_{N-2}(b_{N-2} - b_{N-1})/b_{N-1} \geq 1$. (2) Since $p_i(j+1) = q_{i+j+1} - q_i$, the definitions give $q_i h_i(0) + h_{i+1}(0) = 1$, which is the normalised division equation; and $h_{N-1}(0) = 1/q_{N-1} = b_{N-1}/a$; since $\mu_{N-1} = \delta_0$, also $h_{N-1}(s) = 0$ for $s \geq 1$. (3) Use $p_i(j+1) = k_i + p_{i+1}(j)$ and the binomial theorem. (4) For $i = N - 1$ this holds with $z = -(-q_{N-1})^s b_{N-1}/a = -(-1)^s q_{N-1}^{s-1}$ for $s \geq 1$, since $q_{N-1} b_{N-1} = a$. Then induct backwards on i using (3), $b_{i+1}/a = 1 - q_i b_i/a$ and $k_i - q_{i+1} = -q_i$. □

Theorem 10.2. *For* $1 \leq r \leq N - i$, *the number* $I_r(i) = a \det\big(h_i(j+l)\big)_{0 \leq j,l < r}$ *is a positive integer. Moreover* $I_1(i) = b_i$, $I_2(i) = d_i$ *(if* $i+1 < N$*),* $I_3(i) = D_i$ *(if* $i+3 < N$*), and* $I_4(i) = L_i$ *(if* $i+5 < N$*), where* D_i *and* L_i *are the curvature and the six-step integer of the blocks starting at* b_i.

Proof. Integrality. By Lemma 10.1(4), $h_i(s+1) + q_i h_i(s) \in \mathbb{Z}$. Multiply the Hankel matrix on the left by the unit lower triangular matrix that adds q_i times row $j-1$ to row j. This makes rows $1, \ldots, r-1$ integral and leaves the determinant unchanged. Expand along row 0, whose entries are $(-q_i)^l b_i/a + (\text{integer})$; then $a \det$ is an integer.

Positivity. The Hankel matrix is the Gram matrix of $1, x, \ldots, x^{r-1}$ in $L^2(\nu_i)$. Its quadratic form is $\sum_j w_i(j) q_{i+j}^{-1} \big(\sum_l c_l p_i(j)^l\big)^2$. If this vanishes, the polynomial $\sum_l c_l x^l$ vanishes at the $N - i \geq r$ distinct points $p_i(j)$, so it is zero, by the invertibility of the Vandermonde matrix. So the matrix is positive definite.

The first four. By Lemma 10.1(2),(3), the entries $h_i(s)$, $s \leq 2r - 2$, are explicit polynomials in $\beta_j = b_{i+j}/a$ and $k_i, \ldots, k_{i+2r-4}$; at the end of the orbit this uses $h_{N-1}(s) = 0$ for $s \geq 1$, that is, $b_N = 0$. The claimed values are linear in the b_{i+j}. So each identity holds modulo the relations $q_{i+j}\beta_j + \beta_{j+1} = 1$, with explicit certificates of at most 156 terms per coefficient. □

The ranges $i + 3 < N$ and $i + 5 < N$ cannot be relaxed to all i at which D_i or L_i is defined. For $a = 926$ and $b = 229$ the orbit is $229, 10, 6, 2$, so $N = 4$, and $I_3(1) = 984$ while $D_1 = 988$.

Remark 10.3. So the positivity of D and L, proved earlier by explicit identities, is an instance of the positivity of Gram determinants. The integrality of L (Lemma 7.1(1)) is an instance of the integrality of I_4. A priori $I_r(i)$ involves the moments $h_i(s)$ for $s \leq 2r - 2$, and hence $b_i, \ldots, b_{i+2r-2}$. In fact $I_3(i)$ depends only on b_i, b_{i+1}, b_{i+2} and q_i, q_{i+1}, q_{i+2}. Similarly $I_4(i)$ depends only on $b_i, \ldots, b_{i+3}$ and $q_i, \ldots, q_{i+4}$: the later remainders cancel.

11. Breaking the exponent 3/10

In this section we prove Theorem 1.4. We keep the notation of Section 10: $b_0 > b_1 > \cdots > b_{N-1} > 0$ is the positive part of the orbit, $N = P(a,b)$, and $q_i = \lfloor a/b_i \rfloor$.

11.1. **Heuristics.** The exponent 3/10 of Theorem 1.1 comes from a single balanced configuration: $T \asymp a^{3/10}$ remainders in a window $[A, 2A)$ with $A \asymp a^{3/5}$, quotients $q \asymp a^{2/5}$ and typical increments $k \asymp a^{1/10}$. At this scale the four-term Hankel invariant $I_4(i) = L(i)$ has size $\asymp aH^6/q^4 \asymp 1$, where H bounds the increments. The idea is that a positive integer of bounded size carries information. The labels $I_4(i), I_4(i+1), I_4(i+2)$, together with four further integers of the same size, determine a *pattern*. States with a given pattern satisfy two linear equations and one quadratic equation in five consecutive increments. So a pattern accounts for at most $O(H^2)$ increment tuples, and each tuple for at most four states. At the balanced point this gives $T \ll H^2 \asymp a^{1/5}$, a large saving. Moving away from the balanced point costs a power of the pattern size, and optimising gives the exponent 23/77.

11.2. **The pattern matrix.** Fix a state i with $i + 6 < N$ and write, for $m \geq 0$,

$$k_m = q_{i+m+1} - q_{i+m}, \qquad d_m = b_{i+m} - b_{i+m+1},$$
$$s_m = k_0 + \cdots + k_{m-1}, \qquad r_m = k_m(s_m + s_{m+1}) + k_{m+1}.$$

So $s_m = q_{i+m} - q_i$, $r_0 = k_0^2 + k_1$, $r_1 = k_1(2k_0 + k_1) + k_2$, and so on. Consider the 3×4 integer matrix

$$\mathcal{M}_i = \begin{pmatrix} d_0 & d_1 & d_2 & d_3 \\ k_0 & k_1 & k_2 & k_3 \\ r_0 & r_1 & r_2 & r_3 \end{pmatrix},$$

and let $U_m(i) = (-1)^{m+1} \det(\mathcal{M}_i \text{ with column } m \text{ deleted})$ for $0 \leq m \leq 3$.

Lemma 11.1. (1) *$U_3(i) = L(i) = I_4(i)$ and $U_0(i) = -L(i+1) = -I_4(i+1)$, where L is the six-step integer of Lemma 7.1.*
(2) *$\sum_m U_m(i)\, d_m = \sum_m U_m(i)\, k_m = \sum_m U_m(i)\, r_m = 0$.*

Proof. (2) is the Laplace expansion of a 4×4 determinant with a repeated row. For (1), the identity $\det(\text{columns } 0,1,2) = L(i)$ holds as a polynomial identity in $d_0, d_1, d_2, k_0, \ldots, k_3$, by direct expansion; and $L(i) = I_4(i)$ by Theorem 10.2. Passing from i to $i+1$ shifts the d- and k-rows by one column, and, since $s_m(i+1) = s_{m+1}(i) - k_0$, it replaces r_m by $r_{m+1} - 2k_0k_{m+1}$. This row operation does not change the determinant, so $\det(\text{columns } 1,2,3) = L(i+1)$. □

Remark 11.2. The vector $C_m = U_m/U_3$ has a meaning: $\pi_i = \sum_{m \leq 3} C_m(F_{m+1} - F_m)$, with Newton polynomials $F_m(x) = \prod_{l<m}(x - s_l)$, is the monic orthogonal polynomial of degree four for the measure ν_i of Section 10. The three rows of $\mathcal{M}_i$ encode $\nu_i(\pi_i) = 0$ and $\mu_i(x\pi_i) = \mu_i(x^2\pi_i) = 0$, and $\nu_i(xf) = \mu_i(f) - q_i\nu_i(f)$ turns these into $\nu_i(x^j\pi_i) = 0$ for $j \leq 3$. In particular $U_m(i)/I_4(i)$ are tail sums of the Newton coefficients of π_i: writing $\pi_i = \sum_{l \leq 4} \gamma_l F_l$, one has $U_m(i)/I_4(i) = \gamma_{m+1} + \cdots + \gamma_4$, and $\gamma_0 = -C_0 = I_4(i+1)/I_4(i)$. None of this is used below.

The *pattern* of i is the vector of seven integers

$$\theta(i) = \big(L(i),\, L(i+1),\, L(i+2),\, U_1(i),\, U_2(i),\, U_1(i+1),\, U_2(i+1)\big),$$

and its *gaps* are $\kappa(i) = (k_0, \ldots, k_4)$. By Lemma 11.1, $\theta(i)$ determines $U(i)$ and $U(i+1)$ completely.

11.3. **Three consequences.**

Lemma 11.3 (The reduced quadratic)**.** *Write* $\theta(i) = (N_0, N_1, N_2, u_1, u_2, v_1, v_2)$ *and* $\kappa(i) = (k_0, \dots, k_4)$*. Then*

$$N_1 N_0 (N_1 + N_0)\, k_3^2 + \beta\, k_3 + \gamma = 0,$$

where β *and* γ *are polynomials in* $\theta(i)$*,* k_1 *and* k_2 *alone.*

Proof. Lemma 11.1(2) for the k-row at i and at $i+1$ gives

$$N_1 k_0 = u_1 k_1 + u_2 k_2 + N_0 k_3, \qquad N_1 k_4 = N_2 k_1 - v_1 k_2 - v_2 k_3.$$

Substitute these into $N_1^2 \sum_m U_m(i) r_m = 0$. Since k_0 enters the r_m at most quadratically and k_4 only linearly, the result is a polynomial in (θ, k_1, k_2, k_3). The coefficient of k_3^2 comes from $-N_1 \cdot N_1^2 k_0^2$, from $N_0 \cdot N_1^2 \cdot 2k_0 k_3$ and from $N_0 \cdot N_1^2 k_3^2$, and it equals $-N_1 N_0^2 + 2N_1 N_0^2 + N_1^2 N_0 = N_1 N_0 (N_1 + N_0)$. Explicitly $\beta = 2N_1^2 N_0 (k_1 + k_2) + N_1^2 u_2 + 2N_1 N_0 (u_1 k_1 + u_2 k_2) - N_1 N_0 v_2$, and γ is given in the Lean file `TwoState.lean`. □

Lemma 11.4 (The multiplicity identity)**.** *Let* $F_m(Y) = \prod_{l<m} (Y - s_l)$ *and* $\mathcal{Q}_i(Y) = \sum_{m=0}^{3} U_m(i)\, F_m(Y)\, (Y - s_m - 1)$*. Then*

$$\mathcal{Q}_i(-q_i) = a\, H_4(k_0, \dots, k_4),$$

where H_4 *is an explicit polynomial with* 43 *terms, independent of* i*. As a polynomial in* q*,* $\mathcal{Q}_i(-q)$ *has degree four and leading coefficient* $U_3(i) = L(i) \geq 1$*.*

Proof. The division equations read $b_{i+m+1} = a - (q_i + s_m) b_{i+m}$. Use them to eliminate $b_{i+1}, \dots, b_{i+4}$ from both sides. The difference becomes the zero polynomial in $a, q_i, b_i, k_0, \dots, k_4$; explicit certificates are in the Lean proof. The polynomial H_4 is the Hankel determinant $\det\big(\mu_i(x^{j+l})\big)_{j,l<4}$, and the identity is $I_4(i)\, \pi_i(-q_i) = a \det\big(\mu_i(x^{j+l})\big)$. This follows from the determinant formula for π_i, since $\nu_i(x^{s+1}) + q_i \nu_i(x^s) = \mu_i(x^s)$; we do not need this interpretation. The leading coefficient comes from the term $m = 3$. □

Lemma 11.5 (Labels)**.** *Let* $i + 6 < N$ *and put*

$$G(k_0, \dots, k_4) = (k_0 + 1)\big(k_1 s_2 + s_2 + k_1 + k_2 + 1\big)\big(k_2 (k_1 + k_2) s_3 + e_3\big),$$

where $e_3 = s_3(k_1 + k_2) + s_3 k_2 + (k_1 + k_2) k_2 + (k_0 + 2k_1 + 3k_2)(k_3 + 1) + k_3^2 + 2k_3 + k_4 + 1$*. Then* $L(i)\, q_i^4 \leq a\, G(k_0, \dots, k_4)$*. If moreover* $b_i < 2A$*, then* $L(i)\, a^3 \leq 256 A^4\, G(k_0, \dots, k_4)$*.*

Proof. The Newton polynomials $F_0, \dots, F_3$ are monic of degrees $0, \dots, 3$. So the matrix $\big(\nu_i(F_j F_l)\big)$ is $\mathcal{U}\mathcal{H}\mathcal{U}^{\mathsf{T}}$, where $\mathcal{H}$ is the Hankel matrix of ν_i and $\mathcal{U}$ is unit lower triangular. Hence it has determinant $I_4(i)/a$, and it is positive definite, as in the proof of Theorem 10.2. By Hadamard's inequality,

$$\frac{I_4(i)}{a} \leq \prod_{j<4} \nu_i(F_j^2) \leq \prod_{j<4} \frac{\mu_i(F_j^2)}{q_i},$$

since $\nu_i = \mu_i/(q_i + x) \leq \mu_i/q_i$ on the nonnegative support. By the shift rule $\mu_i(F_j g) = \mu_{i+j}\big(g(x + s_j)\big)$ and the moments $\mu_{i+m}(1) = \mu_{i+m}(x) = 1$, $\mu_{i+m}(x^2) = k_m + 1$, $\mu_{i+m}(x^3) = k_m^2 + 2k_m + k_{m+1} + 1$, the product equals G. For the second claim, $a = q_i b_i + b_{i+1} < 2A(q_i + 1) \leq 4A q_i$, so $L(i) a^4 \leq 256 A^4 L(i) q_i^4 \leq 256 A^4 a G$. □

11.4. **The size of a pattern.**

Lemma 11.6. *Let $1 \le k_0, \dots, k_5 \le H$ and $A \ge 1$. Suppose $N_0 = L(i)$ and $N_1 = L(i+1)$ satisfy $N_0 a^3 \le 256A^4 G(k_0, \dots, k_4)$ and $N_1 a^3 \le 256A^4 G(k_1, \dots, k_5)$, and that the* scale condition

$$237568\, A^4 H^5 \le a^3 \tag{11}$$

holds. Then $L(i)$, $L(i+1)$, $|U_1(i)|$ and $|U_2(i)|$ are all at most $172032\, A^4 H^6/a^3$.

Proof. Put $V = k_1 k_2 (k_1 + k_2)$ and $\Delta_4 = k_0 \cdot k_1 s_2 \cdot k_2 (k_1 + k_2) s_3$. Expanding G and bounding each of the three error terms by $56H^5$, $170H^5$ and $238H^5$ gives $G(k_0, \dots, k_4) \le \Delta_4 + 464H^5$. Also $\Delta_4 \le 6H^3 V$, because $\Delta_4 = k_0 s_2 s_3 \cdot V$. Similarly $G(k_1, \dots, k_5) \le 6H^3 V + 464H^5$, because the corresponding product is $V \cdot k_3 (k_2 + k_3)(k_1 + k_2 + k_3)$. By (11) and $N_0, N_1 \ge 1$, $256A^4 \cdot 464H^5 \le a^3/2 \le N_j a^3/2$ for $j = 0, 1$, so

$$N_0 a^3,\ N_1 a^3 \le 3072\, A^4 H^3 V.$$

Since $N_0 \ge 1$, this and (11) give $V \ge 77H^2$.

Now solve for $(u_1, u_2) = (U_1(i), U_2(i))$ from the k- and r-relations of Lemma 11.1. These read $u_1 k_1 + u_2 k_2 = N_1 k_0 - N_0 k_3$ and $u_1 r_1 + u_2 r_2 = N_1 r_0 - N_0 r_3$. The determinant is

$$\delta = k_1 r_2 - k_2 r_1 = k_1 k_2 (k_1 + k_2) + k_1 k_3 - k_2^2 \ \ge\ V - H^2 \ \ge\ V/2.$$

Using $r_m \le (2m+2)H^2$, Cramer's rule gives $|u_j|\delta \le 14(N_0 + N_1)H^3$, hence

$$|u_j|\, V a^3 \le 28H^3 (N_0 + N_1) a^3 \le 28H^3 \cdot 6144 A^4 H^3 V,$$

that is, $|u_j| a^3 \le 172032 A^4 H^6$. Finally $N_j a^3 \le 3072 A^4 H^3 V \le 6144 A^4 H^6$, since $V \le 2H^3$. □

The point of the argument is in the first paragraph. The two adjacent labels share the Vandermonde factor V, so the ratio $L(i+1)/L(i)$ never enters the bound.

11.5. **The counting lemma.**

Proposition 11.7. *Let $\mathcal{G}$ be a set of states i with $i + 8 < N$. Suppose that for each $i \in \mathcal{G}$ all entries of $\theta(i)$ have absolute value at most M, and all entries of $\kappa(i)$ lie in $[1, H]$. Then $|\mathcal{G}| \le 8(2M+1)^7 H^2$.*

Proof. (i) At most four states per (θ, κ). If $\theta(i) = \theta$ and $\kappa(i) = \kappa$, then by Lemma 11.4 q_i is a root of the polynomial $\mathcal{Q}(-q) - aH_4(\kappa)$. This polynomial depends only on (θ, κ), and it has degree four and leading coefficient $\theta_0 \ge 1$. Since q_i is strictly increasing in i, there are at most four such states.

(ii) At most two values of κ per (θ, k_1, k_2). By Lemma 11.3, k_3 is a root of a quadratic whose coefficients depend only on (θ, k_1, k_2). The leading coefficient is $N_1 N_0 (N_1 + N_0) > 0$. The two linear relations then determine k_0 and k_4, because $N_1 \ne 0$.

(iii) There are at most $(2M+1)^7 H^2$ values of (θ, k_1, k_2). □

11.6. **The window bound.**

Proposition 11.8. *Let $b_s, \dots, b_{s+T-1}$ be a window run in $[A, 2A)$ with $T \ge 36$, and put $H = \lfloor 28a/(AT) \rfloor + 1$ and $M = \lfloor 172032 A^4 H^6/a^3 \rfloor$. If (11) holds, then*

$$T \le 16\, (2M+1)^7 H^2.$$

Proof. Call a start i *good* if $s \le i$, $i + 9 \le s + T$ and $q_{j+1} - q_j \le H$ for $i \le j \le i + 6$, that is, $k_0, \dots, k_6 \le H$ at state i.

Most starts are good. Let $\mathcal{B}$ be the set of $j \in [s, s+T-1)$ with $q_{j+1} - q_j > H$. Then $|\mathcal{B}|(H+1) \le q_{s+T-1} - q_s \le a/A$. Since $H \cdot AT > 28a$, this gives $28|\mathcal{B}| \le T$. Each $j \in \mathcal{B}$ spoils at most seven starts, so at least $T - 9 - 7|\mathcal{B}| \ge T/2$ starts are good, as $T \ge 36$.

Good starts satisfy Proposition 11.7. Let i be good. Then $i+8 < N$, and $b_i, b_{i+1}, b_{i+2} < 2A$. So Lemma 11.5 applies at i, $i+1$ and $i+2$. Lemma 11.6 at i and at $i+1$ bounds all seven entries of $\theta(i)$ by M. Therefore $T/2 \le 8(2M+1)^7 H^2$. □

11.7. **Proof of Theorem 1.4.**

Proposition 11.9. *Let $T \ge 36$ be the length of a window run in $[A, 2A)$. Then $T^{77} \le C_{77}\Delta^{22}a^{23}$, where*

$$C_{77} = 122449^{20} + (8 \cdot 237568 \cdot 56^5)^{11} + (64 \cdot 50176)^{11} + 8^{16} \cdot 16 \cdot 3^7 \cdot 172032^7 \cdot 56^{44}.$$

Proof. We use Proposition 7.3, $T^6A^2 \le 96 \cdot 10^6 a^3$, and Proposition 5.1, which gives $T^2 \le 2T(T-1) \le 8A\Delta$.

Case $AT > 28a$. Then $784a^2T^4 \le T^6A^2 \le 96 \cdot 10^6 a^3$, so $T^4 \le 122449\, a$ and $T^{77} \le 122449^{20}a^{20}$.

From now on $AT \le 28a$, so $H \cdot AT \le 28a + AT \le 56a$.

Case (11) *fails.* Then $a^3(AT)^5 < 237568A^4(H\,AT)^5 \le 237568 \cdot 56^5 A^4a^5$, so $AT^5 \le 237568 \cdot 56^5 a^2$. Hence $T^7 = T^5T^2 \le 8\Delta\, AT^5 \le 8 \cdot 237568 \cdot 56^5\, \Delta a^2$. Raise this to the eleventh power.

Case (11) *holds and $M = 0$.* Then $T \le 16H^2$, so $T(AT)^2 \le 16(56a)^2$. Hence $T^7 = T^3(T^2)^2 \le 64\Delta^2\, T^3A^2 \le 64 \cdot 50176\, \Delta^2 a^2$. This case is in fact vacuous: the proof of Proposition 11.8 provides a good start i, and there $1 \le L(i) \le M$ by Lemma 11.6. We keep it, here and in the analogous proofs below, because the Lean proofs treat it separately.

Case (11) *holds and $M \ge 1$.* Then $2M+1 \le 3M$ and $Ma^3 \le 172032A^4H^6$, so

$$T\, a^{21} \le 16 \cdot 3^7\, (Ma^3)^7 H^2 \le 16 \cdot 3^7 \cdot 172032^7\, A^{28}H^{44}.$$

Multiplying by $(AT)^{44}$ and using $H \cdot AT \le 56a$ gives $T^{45}A^{16} \le 16 \cdot 3^7 \cdot 172032^7 \cdot 56^{44}\, a^{23}$. Finally

$$T^{77} = T^{45}(T^2)^{16} \le 8^{16}\Delta^{16}\, T^{45}A^{16} \le 8^{16} \cdot 16 \cdot 3^7 \cdot 172032^7 \cdot 56^{44}\, \Delta^{16}a^{23}.$$

In every case $T^{77} \le C_{77}\Delta^{22}a^{23}$. □

Proof of Theorem 1.4. This is the argument of Theorem 1.1, with Proposition 11.9 in place of Corollary 5.3. Let $\varepsilon > 0$ and choose an integer $m > 1/\varepsilon$. By Lemma 2.2, $\Delta^{22m} \le C_{22m} \cdot 32a^7$, so every dyadic window has $T \le 35$ or

$$T^{77m} \le C_{77}^m\, C_{22m}\, 32\; a^{23m+7}.$$

Since $(23m+7)/(77m) = 23/77 + 1/(11m) \le 23/77 + \varepsilon/2$, each of the $\lfloor \log_2 a \rfloor + 1$ windows has $T \ll_\varepsilon a^{23/77+\varepsilon/2}$, and the theorem follows. □

Remark 11.10. In terms of the scales, Proposition 11.8 says $T \ll X^7H^2$, where $X = A^4H^6/a^3$ bounds the seven pattern integers and $H \asymp a/(AT)$. Thus $X \asymp a^3/(A^2T^6)$ and $T^{45}A^{16} \ll a^{23}$. The drop bound $A \gg T^2/\Delta$ then gives $T^{77} \ll \Delta^{16}a^{23}$. The scale condition (11) says $Q^4 \gg aH^5$, where $Q = a/A$, and its failure gives the stronger bound $T \ll a^{2/7+o(1)}$.

12. The ternary quadratic and the exponent 11/37

Proposition 11.7 uses only one coefficient of the reduced quadratic. Here we use all of it, which saves a factor H in the count per pattern and proves Theorem 1.5. We keep the notation of Section 11, and, as in Lemma 11.3, write a pattern as $\theta = (N_0, N_1, N_2, u_1, u_2, v_1, v_2) = (L(i), L(i+1), L(i+2), U_1(i), U_2(i), U_1(i+1), U_2(i+1))$. In particular A is still the scale of the window $[A, 2A)$.

12.1. **Signs.** Put $S(a,b,c) = a^2b+(a-1)b^2+ac$ and $E(a,b,c,d) = a^2c+(2a-1)bc+ac^2+ad$. Both are positive at positive integers.

Lemma 12.1 (Signs)**.** *If $N_0, N_1 \geq 1$ and $k_0, \dots, k_4 \geq 1$, then $u_1 > 0 > u_2$.*

Proof. Solving the k- and r-relations of Lemma 11.1 for (u_1, u_2) by Cramer's rule gives the exact identities

$$u_1 S(k_1,k_2,k_3) = N_1 E(k_0,k_1,k_2,k_3) + N_0 S(k_2,k_3,k_4),$$
$$-u_2 S(k_1,k_2,k_3) = N_1 S(k_0,k_1,k_2) + N_0 E(k_1,k_2,k_3,k_4),$$

since the Cramer determinant is $k_1 r_2 - k_2 r_1 = S(k_1,k_2,k_3)$. □

12.2. **The ternary relation.** For a pattern θ let

$$F_\theta(x,y,z) = (u_1x+u_2y+N_0z)^2 + N_1u_1x^2 + N_1u_2y^2 + N_0N_1z^2 + 2N_1u_2xy + 2N_0N_1xz$$
$$+ 2N_0N_1yz + (N_0N_2 - N_1^2)x + (N_1u_1 - N_0v_1)y + (N_1u_2 - N_0v_2)z.$$

Lemma 12.2. *At every state i with $i+7 < N$, $F_{\theta(i)}(k_1,k_2,k_3) = 0$. Write $F_\theta = \alpha x^2 + \beta xy + \gamma y^2 + d(z)x + e(z)y + f(z)$. Then $\alpha = u_1(u_1+N_1)$, $\beta^2 - 4\alpha\gamma = -4N_1u_2(N_1+u_1)(u_1-u_2)$, and*

$$(\beta d_1 - 2\alpha e_1)^2 + (\beta^2 - 4\alpha\gamma)(4\alpha f_2 - d_1^2) = 16\, N_0N_1^2u_1(N_0-u_2)(N_1+u_1)^2(u_1-u_2),$$

where d_1, e_1, f_2 are the leading coefficients of d, e, f. In particular, under the hypotheses of Lemma 12.1, $\alpha > 0$, $\beta^2 - 4\alpha\gamma > 0$ and the last quantity is nonzero.

Proof. The identity $N_1F_\theta(k_1,k_2,k_3) = N_1^2R + (N_0N_1k_3 - N_1^2k_0 + N_1u_1k_1 + N_1u_2k_2)K_0 - N_0N_1K_1$, where K_0, K_1 and R are the k-relations at i, $i+1$ and the r-relation at i, gives the first claim. The rest is expansion. So $F_\theta \equiv N_1R$ modulo the two linear relations. □

12.3. **Norm equations.**

Lemma 12.3 (Norm equations)**.** *Let $D \geq 1$, $m \neq 0$, $B \geq 1$ be integers and $Z = B(\lfloor\sqrt{D}\rfloor + 2)$. The number of $(x,y) \in [-B,B]^2$ with $x^2 - Dy^2 = m$ is at most $2\tau(|m|)^3(\log_3 Z^2 + 1)$.*

Proof. If $D = h^2$ then $(x-hy)(x+hy) = m$, and $x - hy$ is one of the $2\tau(|m|)$ signed divisors of m; it determines (x,y). Otherwise write $D = df^2$ with $d > 1$ squarefree, so d is not a square and $\sqrt{d}$ is irrational. For a solution put $g = \gcd(x, fy)$, $u = x/g$, $w = fy/g$ and $n = u^2 - dw^2$, so that $m = g^2n$. Since $\gcd(u,w) = 1$ and $n \equiv u^2 \pmod{w}$, w is invertible modulo n. Let $t \in [0,|n|)$ be the residue of uw^{-1}; then $t^2 \equiv d$ and $u \equiv tw$ modulo n.

The residues. The number of square roots of d modulo $c \geq 1$ is at most $\tau(c)^2$. Modulo p^e it is at most four. If $p \mid d$, either $e = 1$ and the only root is 0, or $e \geq 2$ and a root would force $p^2 \mid d$. If $p \nmid d$ is odd, two roots r, s satisfy $p^e \mid (s-r)(s+r)$, and p cannot divide both factors, so $s \equiv \pm r$. If $p = 2 \nmid d$, then s, r are odd, one of $(s \mp r)/2$ is odd, and $s \equiv \pm r \pmod{2^{e-1}}$. The count is submultiplicative over coprime factors, by the Chinese remainder theorem, and $4 \leq \tau(p^e)^2$.

A class. Fix g (with $g^2 \mid m$), t and the sign of $\xi = u + w\sqrt{d}$. Two solutions $(u,w), (u',w')$ in this class satisfy $n \mid uu' - dww'$ and $n \mid wu' - uw'$. So $\xi = \xi'\varepsilon$ with $\varepsilon = r + s\sqrt{d}$, where $r = (uu' - dww')/n$, $s = (wu' - uw')/n$ and $r^2 - ds^2 = 1$. If $|\xi| > |\xi'|$ then $\varepsilon > 1$, and then $\varepsilon \geq 3$: indeed $\varepsilon^{-1} = r - s\sqrt{d} \in (0,1)$, so $2r = \varepsilon + \varepsilon^{-1} > 2$, hence $r \geq 2$, and $\varepsilon < 3$ would give $\varepsilon + \varepsilon^{-1} < 4$. Now $|u| + |w|\sqrt{d} \leq |x| + |y|\sqrt{D} \leq Z$, so $|\xi|, |\bar{\xi}| \leq Z$, and $|\xi\bar{\xi}| = |n| \geq 1$ gives $Z|\xi| \geq 1$. Hence the map $\xi \mapsto \lfloor \log_3(Z|\xi|) \rfloor$ is injective on the class, with values in $[0, \log_3 Z^2]$.

There are at most $\tau(|m|)$ values of g, $\tau(|m|)^2$ of t and two signs. □

The argument is the classical one; compare Vaughan and Wooley [13, Lemma 3.5].

12.4. **Slices.**

Lemma 12.4 (Slices)**.** *Let $F(x,y,z)=\alpha x^2+\beta xy+\gamma y^2+(d_1z+d_0)x+(e_1z+e_0)y+f_2z^2+f_1z+f_0$ have integer coefficients bounded by K, with $\alpha>0$, $D=\beta^2-4\alpha\gamma>0$ and $\ell=(\beta d_1-2\alpha e_1)^2+D(4\alpha f_2-d_1^2)\neq 0$. Let $H\geq 1$, and suppose that $2\tau(|m|)^3(\log_3\big((11K^2H(3K+2))^2\big)+1)\leq W$ for all integers $0<|m|\leq 116K^4H^2$. Then F has at most $(2H+1)(W+4)$ zeros in $[-H,H]^3$.*

Proof. Put $t=2\alpha x+\beta y+d(z)$, $\Lambda(z)=\beta d(z)-2\alpha e(z)$, $s=Dy+\Lambda(z)$ and $\mathcal{N}(z)=\Lambda(z)^2+D(4\alpha f(z)-d(z)^2)$. Completing squares gives $s^2-Dt^2=\mathcal{N}(z)-4\alpha DF$. In the box, $|s|,|t|\leq 11K^2H$, $\sqrt{D}\leq 3K$ and $|\mathcal{N}(z)|\leq 116K^4H^2$. For fixed z the map $(x,y)\mapsto(s,t)$ is injective. So if $\mathcal{N}(z)\neq 0$ the slice has at most W zeros, by Lemma 12.3. The z^2-coefficient of $\mathcal{N}$ is $\ell\neq 0$, so at most two slices have $\mathcal{N}(z)=0$. In such a slice $s^2=Dt^2$, which for fixed y is a quadratic in x with leading coefficient $4\alpha^2D\neq 0$, so it has at most $2(2H+1)$ zeros. □

12.5. **The count.**

Proposition 12.5. *Under the hypotheses of Proposition 11.7, let $K=4M^2$, and suppose W satisfies the hypothesis of Lemma 12.4. Then $|\mathcal{G}|\leq 4(2M+1)^7(2H+1)(W+4)$.*

Proof. As before, at most four states share (θ,κ), and κ is determined by (θ,k_1,k_2,k_3). For a fixed θ, the triples (k_1,k_2,k_3) are zeros of F_θ in $[1,H]^3$ (Lemma 12.2). The coefficients of F_θ are at most $4M^2$, and Lemmas 12.1 and 12.2 give the nondegeneracy hypotheses of Lemma 12.4. □

Let $P_1(a)=10^{300}a^{100}$, let $\tau^*(X)=\max_{n\leq X}\tau(n)$ with the convention $\tau(0)=0$, and let $\Xi(a)=2\tau^*(P_1(a))^3(\log_3 P_1(a)+1)+4$. In a window $H\leq 29a$ and $M\leq 10^{16}a^7$, also for the larger M of Proposition 13.4 below, so $116K^4H^2\leq P_1(a)$ and $(11K^2H(3K+2))^2\leq P_1(a)$. Thus $W=\Xi(a)-4$ is admissible, and the argument of Proposition 11.8 gives:

Proposition 12.6. *In the setting of Proposition 11.8, if* (11) *holds then $T\leq 24(2M+1)^7H\,\Xi(a)$.*

Proposition 12.7. *For every window run with $T\geq 36$, $T^{74}\leq C_{74}\Delta^{19}\Xi(a)^{19}a^{22}$, where $C_{74}=122449^{19}+(8\cdot 237568\cdot 56^5)^{11}+10752^{19}+8^{15}\cdot 24\cdot 3^7\cdot 172032^7\cdot 56^{43}$.*

Proof. The four cases of Proposition 11.9 carry over. If $AT>28a$, then $T^4\leq 122449a$. If the scale condition fails, then $T^7\leq 8\cdot 237568\cdot 56^5\Delta a^2$. If $M=0$, then $T^2A\leq 1344\,\Xi a$ and $T^4\leq 10752\,\Delta\Xi a$. If $M\geq 1$, then $Ta^{21}\leq 24\cdot 3^7\cdot 172032^7A^{28}H^{43}\Xi$, so $T^{44}A^{15}\leq 24\cdot 3^7\cdot 172032^7\cdot 56^{43}\,\Xi a^{22}$, and $T^{74}=T^{44}(T^2)^{15}\leq(8A\Delta)^{15}T^{44}$. □

Proof of Theorem 1.5. By the divisor bound $\tau(n)^k\leq C_kn$, both Δ and $\tau^*(P_1(a))$ are $O_\eta(a^\eta)$, and $\log_3 P_1(a)\ll_\eta a^\eta$. So $\Xi(a)\ll_\eta a^\eta$, and Proposition 12.7 gives $T\ll_\eta a^{(22+38\eta)/74}$ for every dyadic window. Since $22/74=11/37$, summing over the $O(\log a)$ windows proves the theorem. □

Remark 12.8. In terms of the scales, the count per pattern is now $O(Ha^\eta)$ instead of $O(H^2)$. So $T\ll a^\eta X^7H$, which gives $T^{44}\ll a^{7+\eta}Q^{15}$, and with $Q\ll a^{1+\eta}/T^2$ the bound $T^{74}\ll a^{22+O(\eta)}$. At the critical point $T\approx a^{11/37}$, $Q\approx a^{15/37}$, $H\approx a^{4/37}$ and $X\approx a^{1/37}$.

13. Ten minors and the exponent 8/27

The seven pattern integers of Section 12 were counted as if they were independent. They are not: they are coordinates of a single vector of 3×3 minors. Counting such vectors directly saves a factor X^2 and proves Theorem 1.6.

13.1. **The full minor vector.** Let $\mathcal{M}_i$ now be the 3×5 matrix with rows $(d_0,\dots,d_4)$, $(k_0,\dots,k_4)$ and $(r_0,\dots,r_4)$, and let $p_{c_1c_2c_3}$ be its minor on columns $c_1<c_2<c_3$. Expanding the cofactors of Section 11 gives

$$\theta(i)=(p_{012},\,p_{123},\,p_{234},\,p_{023},\,-p_{013},\,p_{134},\,-p_{124}).$$

Lemma 13.1 (Expansion)**.** *Let $q=q_i$ and $s_j=q_{i+j}-q$. If $i+j+3<N$ then*

$$\frac{b_{i+j}}{a}=\frac{1}{q}-\frac{1+s_j}{q^2}+\frac{s_j^2+2s_j+k_j+1}{q^3}-e_j,\qquad 0\le e_j\le\frac{\mu_{i+j}\big((x+s_j)^3\big)}{q^4}.$$

Proof. Apply μ_{i+j} to $\frac{1}{q+y}=\frac{1}{q}-\frac{y}{q^2}+\frac{y^2}{q^3}-\frac{y^3}{q^3(q+y)}$ with $y=s_j+x\ge 0$, using $b_{i+j}/a=\nu_{i+j}(1)=\mu_{i+j}\big(1/(q+s_j+x)\big)$, $\mu(1)=\mu(x)=1$ and $\mu(x^2)=k+1$. □

Proposition 13.2 (All minors)**.** *Let $i+8<N$, $b_i<2A$ and $k_0,\dots,k_6\le H$ at state i. Then every minor of $\mathcal{M}_i$ satisfies $|p|\,a^3\le 3609600\,A^4H^6$.*

Proof. Subtracting the expansions for j and $j+1$ gives $d_j/a=k_j/q^2-(r_j+k_j)/q^3+e_{j+1}-e_j$. So replacing the first row by $d'_j=d_j-a(q^{-2}-q^{-3})k_j+aq^{-3}r_j=a(e_{j+1}-e_j)$ changes no minor. Now $|d'_j|\le 235aH^3/q^4$, $|k_j|\le H$ and $|r_j|\le 10H^2$, so $|p|\le 14100\,aH^6/q^4$. Finally $a\le 4Aq$. □

This replaces Lemma 11.6, and the scale condition (11) is no longer needed.

13.2. **Counting minor vectors.**

Proposition 13.3 (Minor vectors)**.** *The number of vectors of minors of integer 3×5 matrices with all coordinates in $[-Y,Y]$ is at most $1+20\sum_{m\le Y}729\,m^4\tau(m)\ll Y^{5+o(1)}$.*

Proof. For a nonzero vector p choose a pivot π with $|p_\pi|=|m|$ maximal, and let c_4,c_5 be the two columns outside π. Let u_a (v_a) be the minor with the a-th pivot column replaced by c_4 (c_5), with the sign of the column permutation. By Cramer's rule and the Plücker relations, the remaining three minors satisfy $m\,p_{w_a}=\pm(u\times v)_a$ (thirty polynomial identities, checked symbolically). So p is determined by $(\pi,\operatorname{sgn}m,|m|,u,v)$, where $|u_a|,|v_a|\le|m|$ and $|m|$ divides every coordinate of $u\times v$.

Fix $m\ge 1$ and u, and let $g=\gcd(m,u_1,u_2,u_3)$, $n=m/g$, $u=gu_0$. By Bézout there are $c\in\mathbb{Z}^3$ and $x\in\mathbb{Z}$ with $c\cdot u_0+xn=1$. If $m\mid u\times v$, then $n\mid u_0\times v$, and the identity $v-(c\cdot v)u_0=xn\,v+c\times(v\times u_0)$ shows $v\equiv(c\cdot v)u_0\pmod n$. Hence v lies in one of n classes modulo n, each meeting $[-m,m]^3$ in at most $(3g)^3$ points: at most $27mg^2$ choices. Summing over u, grouped by $g=d\mid m$, gives $\sum_u g^2\le\sum_{d\mid m}d^2(3m/d)^3\le 27m^3\tau(m)$, so at most $729m^4\tau(m)$ pairs (u,v). There are ten pivots and two signs. □

13.3. **The bound.**

Proposition 13.4. *For a window run in $[A,2A)$ with $T\ge 36$, let $H=\lfloor 28a/(AT)\rfloor+1$ and $M=\lfloor 3609600A^4H^6/a^3\rfloor$. Then $T\le 24\cdot 14581\,(M+1)^5H\,\Xi(a)^2$, and*

$$T^{54}\le C_{54}\Delta^{14}\Xi(a)^{28}a^{16}$$

with $C_{54}=122449^{14}+(448\cdot 24\cdot 14581)^{14}+8^{11}\cdot 32\cdot 24\cdot 14581\cdot 3609600^5\cdot 56^{31}$.

Proof. As in Propositions 11.8 and 12.5, at least $T/2$ starts are good, at most four good starts share (θ,κ), and there are at most $(2H+1)(W+4)$ gap triples per pattern. Since θ is a function of the minor vector, there are at most $1+14580M^5\tau^*(M)\le 14581(M+1)^5\Xi(a)$ patterns, by Propositions 13.2 and 13.3. For the second bound, if $AT>28a$ then $T^4\le$

$122449a$. Otherwise $HAT \le 56a$. If $M = 0$ then $T^4 \le 448 \cdot 24 \cdot 14581\,\Delta\Xi^2 a$. Otherwise $Ta^{15} \le 32 \cdot 24 \cdot 14581 \cdot 3609600^5 A^{20} H^{31} \Xi^2$, hence $T^{32}A^{11} \ll \Xi^2 a^{16}$ and $T^{54} \le (8A\Delta)^{11} T^{32}$. □

Proof of Theorem 1.6. Since $\Delta, \Xi(a) \ll_\eta a^\eta$, every window has $T \ll a^{(16+42\eta)/54}$, and $16/54 = 8/27$. □

Remark 13.5. In terms of the scales, $T \ll a^\eta X^5 H$, so $T^{32} \ll a^{5+\eta} Q^{11}$ and $T^{54} \ll a^{16+O(\eta)}$. At the critical point $T \approx a^{8/27}$, $Q \approx a^{11/27}$, $H \approx a^{3/27}$ and $X \approx a^{1/27}$.

14. Lattices of minor vectors and the exponent 45/152

Proposition 13.4 allows $(2H+1)(W+4)$ gap vectors for every minor vector. But the gap vector of a state is not an arbitrary point of $[1,H]^5$: it lies in a rank-three lattice determined by its minor vector. If that lattice has no short vectors, the slices of the ternary quadratic are few. If it has a short vector, the minor vector satisfies an extra congruence, and few minor vectors do. Balancing the two proves Theorem 1.7. The argument uses only integer arithmetic. Instead of the geometry of numbers it uses a reduced basis obtained by minimising determinants and lengths directly.

14.1. **The lattice of a minor vector.** Let $p \neq 0$ be the minor vector of a 3×5 integer matrix $\mathcal{M}$, with pivot π, $m = p_\pi$, pivot columns $c_1 < c_2 < c_3$, other columns $c_4 < c_5$, and u, v as in Proposition 13.3. Let $\Lambda_p \subseteq \mathbb{Z}^3$ be the subgroup of x with $m \mid u \cdot x$ and $m \mid v \cdot x$. It contains $m\mathbb{Z}^3$. Let $\lambda(p)$ be the least sup-norm of a nonzero vector of Λ_p.

Lemma 14.1 (Lifting)**.** *Let L_c be me_a if $c = c_a$ is a pivot column, u if $c = c_4$ and v if $c = c_5$. For every row x of $\mathcal{M}$ and every column c, $m\,x_c = L_c \cdot (x_{c_1}, x_{c_2}, x_{c_3})$. Moreover $\det(L_1, L_2, L_3) = m^2 p_{123}$.*

Proof. The rows L_c are the rows of $\mathcal{M}^{\mathsf{T}} \operatorname{adj}(\mathcal{M}_\pi^{\mathsf{T}})$, where $\mathcal{M}_\pi$ is the pivot 3×3 block. This is Cramer's rule. Both claims are polynomial identities in the fifteen entries, checked symbolically for each of the ten pivots. □

So the pivot gaps $k' = (k_{c_1}, k_{c_2}, k_{c_3})$ of a state lie in Λ_p. Its five gaps are $k_c = L_c \cdot k'/m$, and since $p_{123} = L(i+1) \ge 1$, the gaps (k_1, k_2, k_3) are an invertible linear image of k'.

Lemma 14.2 (Index)**.** *If $(t+1)^3 > |m|$ then $\lambda(p) \le t$.*

Proof. Write $g = \gcd(|m|, u)$, $n = |m|/g$ and $u = gu_0$. The proof of Proposition 13.3 gives $v = \mu u_0 + nw$ with $\mu = c \cdot v \in \mathbb{Z}$ and $w \in \mathbb{Z}^3$. Put $\psi(x) = \big(u_0 \cdot x \bmod n,\ (\mu \lfloor u_0 \cdot x/n \rfloor + w \cdot x) \bmod g\big)$, which takes at most $ng = |m|$ values. Two points $x \neq y$ of $[0,t]^3$ with $\psi(x) = \psi(y)$ exist. Then $u_0 \cdot (x-y) = n(q-q')$ with q, q' the two floors, so $u \cdot (x-y) = \pm m(q-q')$, and $v \cdot (x-y) = n\big(\mu(q-q') + w \cdot (x-y)\big)$ with the bracket divisible by g. So $x - y \in \Lambda_p$. □

Since $\psi(x) = \psi(y)$ forces $x - y \in \Lambda_p$, the same argument shows that Λ_p has index at most $|m|$ in $\mathbb{Z}^3$.

14.2. **Reduced bases.**

Lemma 14.3 (Reduction)**.** *Let $\Lambda \subseteq \mathbb{Z}^3$ be a subgroup containing $m\mathbb{Z}^3$, $m \ge 1$. There is a basis b_1, b_2, b_3 of Λ with $\delta = \det(b_1, b_2, b_3)$, $0 < |\delta| \le m^3$ and*

$$27\,|b_1|^2 |b_2|^2 |b_3|^2 \le 64\,\delta^2.$$

Hence every $w \in \Lambda$ is $\sum y_i b_i$ with $y \in \mathbb{Z}^3$ and $27|b_i|^2 y_i^2 \le 64|w|^2$.

Proof. Let I be the least nonzero $|\det|$ of a triple in Λ; the triple (me_1, me_2, me_3) gives $I \le m^3$. A triple with $|\det| = I$ is a basis. Otherwise some $w \in \Lambda$ has a Cramer coordinate that is not an integer, say the first, $\det(w, b_2, b_3)/\delta$, and $w - \lfloor\cdot\rfloor b_1$ gives a triple with smaller nonzero determinant. Among such bases minimise $d_1 = |b_1|^2$, and then the Gram determinant $d_2 = d_1|b_2|^2 - \langle b_1, b_2\rangle^2$. Put $d_3 = \delta^2$.

Choose n with $|2(\langle b_1, b_2\rangle + nd_1)| \le d_1$. Then $(b_2 + nb_1, b_1, b_3)$ is a basis, so $d_1 \le |b_2 + nb_1|^2$, and $d_1|b_2 + nb_1|^2 = d_2 + (\langle b_1, b_2\rangle + nd_1)^2$. Hence $3d_1^2 \le 4d_2$.

Let $B(y, z) = d_1\langle y, z\rangle - \langle b_1, y\rangle\langle b_1, z\rangle$, so that $B(b_2, b_2) = d_2$, and Sylvester's identity gives $B(b_2, b_2)B(b_3, b_3) - B(b_2, b_3)^2 = d_1 d_3$. Choose n_2 with $|2(B(b_2, b_3) + n_2 d_2)| \le d_2$ and put $c = b_3 + n_2 b_2$. Then (b_1, c, b_2) is a basis with first vector b_1, so $d_2 \le d_1|c|^2 - \langle b_1, c\rangle^2 = B(c, c)$, while $d_2 B(c, c) = (B(b_2, b_3) + n_2 d_2)^2 + d_1 d_3$. Hence $3d_2^2 \le 4d_1 d_3$.

Finally choose n_1 with $|2(\langle b_1, c\rangle + n_1 d_1)| \le d_1$, and use the basis b_1, $b_2' = b_2 + nb_1$, $b_3' = c + n_1 b_1$. The identities above give $3d_1|b_2'|^2 \le 4d_2$ and $4d_1 d_2|b_3'|^2 \le d_2^2 + 4d_1 d_3 + d_1^2 d_2$. With the two inequalities this gives $9d_2|b_3'|^2 \le 16 d_3$, and the product of the three bounds is the claim. For the coordinates, $y_1\delta = \det(w, b_2, b_3)$ and Hadamard's inequality $\det(w, b_2, b_3)^2 \le |w|^2|b_2|^2|b_3|^2$. □

14.3. **Few minor vectors have a short vector.**

Lemma 14.4. *For $n \ge 1$ and $x \in \mathbb{Z}^3$, at most $27n^2 \gcd(n, x)$ vectors $w \in [-n, n]^3$ have $n \mid w \cdot x$.*

Proof. Reducing modulo n loses a factor at most 27. Let K be the set of solutions in $[0, n)^3$, $e = \gcd(n, x)$ and $c\cdot x \equiv e \pmod n$ by Bézout. The map $(w, j) \mapsto (w + jc) \bmod n$ is injective on $K \times [0, n/e)$: equal images give $n \mid (j - j')e$, so $j = j'$, and then $w = w'$. Hence $|K|\, n/e \le n^3$. □

Proposition 14.5 (Short vectors)**.** *Let $\mathcal{S}$ be a set of nonzero minor vectors with coordinates in $[-Y, Y]$, and let $\tau(m) \le \tau_0$ for $m \le Y$. Then for every $R \ge 1$,*

$$\#\{p \in \mathcal{S} : \lambda(p) \le R\} \le 393660\, R^3 Y^4 \tau_0^2.$$

Proof. If $\lambda(p) \le R$, some nonzero $x \in [-R, R]^3$ has $m \mid u \cdot x$. For fixed x, the encoding of Proposition 13.3 gives at most $20 \sum_m N(m, x)$ such p, where $N(m, x)$ counts pairs (u, v) with $m \mid u \times v$ and $m \mid u \cdot x$. For each u there are at most $27m \gcd(m, u)^2$ vectors v. Grouping the u by $g = \gcd(m, u)$ and writing $u = gu'$ with $(m/g) \mid u' \cdot x$, Lemma 14.4 gives $\sum_u g^2 \le 27m^2\tau(m)\gcd(m, x)$, so $N(m, x) \le 729 m^3 \tau(m) \gcd(m, x)$. Finally $\sum_{0<|x|_\infty \le R} \gcd(m, x) \le 27R^3\tau(m)$. A divisor $h > R$ of m divides no nonzero x in the box, and for $h \le R$ the multiples of h number at most $(2R/h + 1)^3$, with $h(2R/h + 1) \le 3R$. □

14.4. **One minor vector.**

Lemma 14.6 (Definite norm equations)**.** *For $D \ge 1$ and $m \ne 0$, the equation $x^2 + Dy^2 = m$ has at most $4\tau(|m|)^3$ integer solutions.*

Proof. As in Lemma 12.3, with $D = df^2$, d squarefree, each solution reduces to a primitive $u^2 + dw^2 = n$, $m = g^2 n$, with a class $t \equiv uw^{-1}$, a square root of $-d$ modulo n. The count of square roots in the proof of Lemma 12.3 applies to $-d$ unchanged. Two solutions in a class give integers r, s with $r^2 + ds^2 = 1$, of which there are at most four. □

A ternary quadratic F with quadratic part of Gram matrix S has $\det(2S)$ multiplied by $\det(C)^2$ under a substitution $x = Cy$. In the notation of Lemma 12.4 one has the identity $\ell = -2\alpha \det(2S)$. For F_θ, Lemma 12.2 therefore gives $\det(2S) = -8N_0N_1^2(N_0 - u_2)(N_1 + u_1)(u_1 - u_2) \ne 0$ under the hypotheses of Lemma 12.1.

If $\det(2S) \ne 0$, a product of two substitutions, each with entries in $\{0, \pm 1\}$, determinant ± 1 and an inverse with absolute row sums at most 2, makes both the y_1^2 coefficient α and the

binary discriminant in (y_1, y_2) nonzero, and then $\ell \neq 0$ as well. First choose one of the planes $\langle e_1, e_2\rangle$, $\langle e_1, e_3\rangle$, $\langle e_2, e_3\rangle$, $\langle e_1 + e_2, e_3\rangle$: if all four binary discriminants vanished then $\det(2S)$ would vanish, by an explicit combination with polynomial coefficients. Then choose f, g or $f + g$ in the plane. With this, Lemma 12.4 holds for $\alpha \neq 0$ and $D \neq 0$ of either sign, under the same hypothesis on W. When $D < 0$ each slice uses Lemma 14.6, whose bound $4\tau(|m|)^3$ is at most $2\tau(|m|)^3(\log_3\big((11K^2H(3K+2))^2\big) + 1)$ because $11K^2H(3K+2) \geq 3$.

Proposition 14.7 (One minor vector)**.** *Let p be a minor vector with all coordinates in $[-X, X]$, $X \geq 1$, and suppose W satisfies the hypothesis of Lemma 12.4 for $K = 248832X^8$ and $12H+1$ in place of H. The good starts with minor vector p have at most*

$$\big(8\lfloor 3H/\lambda(p)\rfloor + 3\big)(W + 4)$$

gap vectors κ.

Proof. At a good start the gaps lie in $[1, H]$, and since $N_0 = L(i)$ and $N_1 = L(i+1)$ are positive integers, Lemma 12.1 gives $u_1 > 0 > u_2$. Let b_1, b_2, b_3 be the basis of Lemma 14.3 for Λ_p, with $|m|$ in place of m, and $\hat{b}_i$ their lifts $c \mapsto L_c \cdot b_i/m$ to $\mathbb{Z}^5$. Write $k' = \sum y_i b_i$. By Lemma 14.1, $\kappa = \sum y_i \hat{b}_i$, so $(k_1, k_2, k_3) = \hat{B}y$ for the 3×3 matrix $\hat{B}$ of the lifts in columns $1, 2, 3$, and $m \det \hat{B} = p_{123}\delta \neq 0$. So y is a zero of $\tilde{F} = F_\theta \circ \hat{B}$, and $\det(2S_{\tilde{F}}) = \det(2S_{F_\theta})(\det \hat{B})^2 \neq 0$. Each $|b_i|^2 \geq \lambda(p)^2$ and $|k'|^2 \leq 3H^2$, so $\lambda(p)|y_i| \leq 3H$, and after the plane substitution the coordinates lie in $[-Y', Y']$ with $Y' = 4\lfloor 3H/\lambda(p)\rfloor + 1 \leq 12H + 1$. Since $|b_i| \leq 2X^3$ by Lemma 14.3, every lift is at most $6X^3$, and the coefficients of the final quadratic are at most $12^3 \cdot 4X^2 \cdot (6X^3)^2 = 248832X^8$. The slice count applies with $(2Y'+1)(W+4)$, and y determines κ. □

14.5. **The bound.**

Lemma 14.8 (Partial summation)**.** *Let t and R_p for $p \in \mathcal{S}$ be integers with $1 \leq R_p \leq t$, and let $\#\{p : R_p \leq r\} \leq Cr^3$ for $r \geq 1$. Then $t\sum_p \lfloor n/R_p\rfloor \leq n|\mathcal{S}| + nCt^3$ for every $n \geq 0$.*

Proof. Write $1/r = 1/t + \sum_{r \leq j < t} 1/(j(j+1))$ and exchange the sums: $\sum_p 1/R_p \leq |\mathcal{S}|/t + \sum_{j<t} Cj^3/(j(j+1)) \leq |\mathcal{S}|/t + Ct^2$. □

Let $P_2(a) = 10^{1000}a^{400}$ and $\Xi_2(a) = 2\tau^*(P_2(a))^3(\log_3 P_2(a) + 1) + 4$. As in Section 12, $W = \Xi_2(a) - 4$ is admissible in every window, and $\Xi_2(a) \ll_\eta a^\eta$.

Proposition 14.9. *For a window run in $[A, 2A)$ with $T \geq 36$, let $H = \lfloor 28a/(AT)\rfloor + 1$ and $X = \lfloor 3609600A^4H^6/a^3\rfloor + 1$. Then*

$$T^3 \leq C_\star\, \Xi_2(a)^9\,(H^3X^{14} + X^{15}), \qquad C_\star = 32 \cdot 607461312^3,$$

and $T^{152} \leq C_{152}\Delta^{39}\Xi_2(a)^{117}a^{45}$ with $C_{152} = 122449^{38} + c_0^{13} + 8^{31}c_A + 8^{30}c_B$, where $c_0 = 1024\, C_\star\, 56^3$, $c_A = 2^{15}C_\star\, 3609600^{14}56^{87}$ and $c_B = 2^{16}C_\star\, 3609600^{15}56^{90}$.

Proof. Let $\mathcal{S}$ be the set of minor vectors of good starts and $\tau_0 = \tau^*(P_2(a))$. At most four good starts share (p, κ), and $T \leq 2|\mathcal{G}|$, so Proposition 14.7 gives

$$T \leq 8\,\Xi_2(a)\Big(8\sum_{p\in\mathcal{S}}\lfloor 3H/\lambda(p)\rfloor + 3|\mathcal{S}|\Big).$$

Choose t with $X \leq t^3 \leq 8X$. Lemma 14.2 gives $\lambda(p) \leq t$, and Proposition 14.5 and Lemma 14.8 give $t\sum_p\lfloor 3H/\lambda(p)\rfloor \leq 3H|\mathcal{S}| + 3H \cdot 393660X^4\tau_0^2t^3$, with $|\mathcal{S}| \leq 14581X^5\tau_0$. Hence $Tt \leq 607461312\,\Xi_2^3X^5(H+t)$, and cubing, with $(H+t)^3 \leq 4(H^3+t^3)$ and $X \leq t^3 \leq 8X$, gives the first bound.

For the second, if $AT > 28a$ then $T^4 \le 122449a$. Otherwise $H\,AT \le 56a$. If $X = 1$ then $T^6A^3 \le 2C_\star\,56^3\Xi_2^9a^3$, and $T^{12} \le (8A\Delta)^3T^6 \le c_0\Delta^3\Xi_2^9a^3$. If $X \ge 2$, write $X = X_0 + 1 \le 2X_0$ with $X_0a^3 \le 3609600A^4H^6$. When $2X_0 \le H^3$, $T^3a^{42} \ll \Xi_2^9A^{56}H^{87}$, so $T^{90}A^{31} \le c_A\Xi_2^9a^{45}$ and $T^{152} \le (8A\Delta)^{31}T^{90}$. When $2X_0 > H^3$, $T^3a^{45} \ll \Xi_2^9A^{60}H^{90}$, so $T^{93}A^{30} \le c_B\Xi_2^9a^{45}$ and $T^{153} \le (8A\Delta)^{30}T^{93}$. □

Proof of Theorem 1.7. Since $\Delta, \Xi_2(a) \ll_\eta a^\eta$, every window has $T \ll a^{(45+156\eta)/152}$. Summing over the $O(\log a)$ windows proves the theorem. □

Remark 14.10. In terms of the scales, $T \ll a^\eta(HX^{14/3} + X^5)$: the factor $X^{1/3}$ saved over Proposition 13.4 is the typical $\lambda(p)$. At the critical point $T \approx a^{45/152}$, $Q \approx a^{62/152}$, $H \approx a^{17/152}$ and $X \approx a^{6/152}$, and the term $X^5 \approx a^{30/152}$ is smaller.

15. The Lean formalisation

The repository accompanying this paper contains a Lean 4 project (toolchain `v4.35.0`, Mathlib `v4.35.0`) that proves the theorems of Section 1, with (9) and (10) as hypotheses for Theorem 1.2. Theorem 1.1 is

```
theorem P_le : ∀ ε > (0 : ℝ), ∃ C > 0,
    ∀ a b : ℕ, 0 < b → b < a →
    (P a b : ℝ) ≤ C * (a : ℝ) ^ ((3 : ℝ) / 10 + ε)
```

and Theorem 1.4 is

```
theorem P_le_23_77 : ∀ ε > (0 : ℝ), ∃ C > 0,
    ∀ a b : ℕ, 0 < b → b < a →
    (P a b : ℝ) ≤ C * (a : ℝ) ^ ((23 : ℝ) / 77 + ε)
```

and Theorem 1.5 is

```
theorem P_le_11_37 : ∀ ε > (0 : ℝ), ∃ C > 0,
    ∀ a b : ℕ, 0 < b → b < a →
    (P a b : ℝ) ≤ C * (a : ℝ) ^ ((11 : ℝ) / 37 + ε)
```

and Theorem 1.6 is

```
theorem P_le_8_27 : ∀ ε > (0 : ℝ), ∃ C > 0,
    ∀ a b : ℕ, 0 < b → b < a →
    (P a b : ℝ) ≤ C * (a : ℝ) ^ ((8 : ℝ) / 27 + ε)
```

and Theorem 1.7 is

```
theorem P_le_45_152 : ∀ ε > (0 : ℝ), ∃ C > 0,
    ∀ a b : ℕ, 0 < b → b < a →
    (P a b : ℝ) ≤ C * (a : ℝ) ^ ((45 : ℝ) / 152 + ε)
```

Here `P a b` is defined by well-founded recursion as the number of positive terms ($P(a,0) = 0$, $P(a,b) = P(a, a \bmod b) + 1$ for $b > 0$), and the file checks $P(35,22) = 7$. Theorem 1.2 is formalised as

```
theorem P_le_loglog (hNT : NairTenenbaumDelta)
    (hBT : BretecheTenenbaumMean) :
    ∃ C > 0,
    ∀ a b : ℕ, 0 < b → b < a →
    (P a b : ℝ) ≤ C * (a : ℝ) ^ (3 / 10 : ℝ)
        * √(Real.log (Real.log (3 * a)))
```

Theorems 1.3 and 10.2 are formalised as

```
theorem P_le_rises : ∃ C > 0,
    ∀ a b : ℕ, 0 < b → b < a →
    (P a b : ℝ) ≤ C * (a : ℝ) ^ (1 / 4 : ℝ) * √(rises a b + 1)
theorem hankelInv_pos_int (hba : b < a) (hi : i < P a b)
    (hr : r ≤ P a b - i) :
    ∃ n : ℕ, 0 < n ∧ hankelInv a b i r = n
```

together with `hankelInv_one`, ..., `hankelInv_four`; the last states `hankelInv a b i 4 = sixL` ..., the six-step integer of Lemma 7.1. Here `NairTenenbaumDelta` and `BretecheTenenbaumMean` are Lean propositions stating (9) (with $\sigma = 1/4$) and (10), with existential constants and thresholds. `hooleyDelta` is defined as the supremum over $u \in \mathbb{R}$, exactly as above. The command `#print axioms` was run on `P_le`, `P_le_23_77`, `P_le_11_37`, `P_le_8_27`, `P_le_45_152`, `norm_count`, `minor_vector_count`, `reduced_basis`, `short_count`, `six_step_bound`, `P_le_of_shortDivisorBound`, `P_le_loglog`, `P_le_rises`, `hankelInv_pos_int` and `hankelInv_four`; for each it reports only `propext`, `Classical.choice` and `Quot.sound`. The file `Axioms.lean` checks this for the eight main theorems with `#guard_msgs`. The proof development contains no `sorry` and no `native_decide`. The separate Mathlib-only `Challenge.lean` states these eight results with deliberate theorem holes for comparison with the proved `Solution.lean`; neither module imports the other. In the tables below, Lean names omit their namespaces (`PierceLengthBound`, `PierceLengthBound.Window`, `OAI.Ostmann`).

Paper	Lean	File
Definition of $P(a,b)$	`P`, `orb`	`Defs.lean`
Lemma 2.1	`div_lt_div_mod`	`Block.lean`
Lemma 3.2	`curvature_identity`, `one_le_bD`	`Block.lean`
Lemma 3.3	`fiber_identity`, `block_fiber_eq`	`Block.lean`
Lemma 3.4	`bu_mul_le`, `bv_mul_le` $(w > 0)$, `bD_mul_le`	`Block.lean`
Lemma 4.1	`fiber_poly_identity`, `fiber_card_le`	`Fiber.lean`
Window runs	`Window`, `DivBound`	`Window.lean`
Proposition 5.1	`drop_bound`	`Window.lean`
Proposition 5.2	`curvature_bound`	`Window.lean`
Corollary 5.3	`window_bound`	`Window.lean`
Dyadic decomposition	`P_eq_sum_visits`, `visits_window`	`Main.lean`
Summing the windows	`SlowGrowth`, `tauMax_slow`, `P_le_of_visits`	`Main.lean`
Theorem 1.1	`P_le`	`Main.lean`
Lemma 2.2	`exists_divisors_power_bound`	`DivisorPowerBound.lean`
Lemma 7.1	`six_divisibility`, `six_identity`	`SixStep.lean`
Corollary 7.2	`six_core`	`SixStep.lean`
	`one_le_bD`	`Block.lean`
	`Lz_pos`	`TwentyThreeMain.lean`
Proposition 7.3	`six_step_bound`	`SixStep.lean`
Proposition 8.1	`drop_count_bound'`	`ShortSum.lean`
Lemma 8.3	`sum_min_le`, `sum_min_le_gen`	`Geometric.lean`
Theorem 8.4	`P_le_of_shortDivisorBound`	`Conditional.lean`
Δ_{H}, $\tau_A \le \Delta_{\mathrm{H}}$	`hooleyDelta`, `dcount_le_hooleyDelta`	`Hooley.lean`
(9), (10)	`NairTenenbaumDelta`, `BretecheTenenbaumMean`	`Hooley.lean`
Lemma 8.5	`sum_hooleyDelta_div_le`	`Hooley.lean`
Theorem 1.2	`P_le_loglog`	`Hooley.lean`
Lemma 9.1	`ap_identity`, `ap_bound`	`Rises.lean`
Lemma 9.3	`ratio_lt`, `drop_lt_of_not_rise`	`Rises.lean`
Lemma 9.4	`sq_card_le_of_fiber`	`Rises.lean`
Proposition 9.5	`rises_drop_bound`, `rises_quot_bound`, `rises_incr_bound`	`Rises.lean`
Theorem 1.3	`P_le_rises`, `P_le_of_no_rises`	`RisesMain.lean`
Lemma 10.1	`wt_spec`, `hm_zero`, `hm_succ`, `hm_int`	`Hankel.lean`
Theorem 10.2	`hankelInv_pos_int`, `hankelInv_one`, …, `hankelInv_four`	`Hankel.lean`

Paper	Lean	File
Hadamard's inequality	`det_le_prod_diag`	`Hadamard.lean`
μ_i, ν_i, shift rule	`muF`, `nuF`, `muF_nb_mul`	`Functional.lean`
Lemma 11.5	`hankelInv_le_prod`, `label_bound`	`Gram.lean`
	`label_at`	`TwentyThreeMain.lean`
Lemma 11.1	`null_d`, `null_k`, `null_r`	`TwoState.lean`
	`theta_eq`, `hankelInv_four`	`EightMain.lean`, `Hankel.lean`
Lemma 11.3	`quad_root`	`TwoState.lean`
Lemma 11.4	`mult_identity`, `quart_eq`	`TwoState.lean`
Lemma 11.6	`gramP_le`, `state_bound`	`PatternBound.lean`
Proposition 11.7	`pattern_count`	`PatternCount.lean`
Proposition 11.8	`good_card`, `two_state_window_77`	`TwentyThreeMain.lean`
Proposition 11.9	`window_77`, `visits_77`	`TwentyThreeMain.lean`
Theorem 1.4	`P_le_23_77`	`TwentyThreeMain.lean`
Lemma 12.1	`pattern_signs`	`ElevenCount.lean`
Lemma 12.2	`tern_rel`, `tern_disc`, `tern_lead`	`ElevenCount.lean`
Lemma 12.3	`card_sqRoots_le`, `class_card`, `norm_count`	`NormCount.lean`
Lemma 12.4	`slice_identity`, `slice_count_ne`, `slice_count`	`SliceCount.lean`
Proposition 12.5	`pattern_count_slice`	`ElevenCount.lean`
Proposition 12.6	`two_state_window_74`	`ElevenMain.lean`
Proposition 12.7	`window_74`, `visits_74`	`ElevenMain.lean`
Theorem 1.5	`xi_le`, `slack_le`, `P_le_11_37`	`ElevenMain.lean`
Lemma 13.1	`nu_expand`	`MinorHeight.lean`
Proposition 13.2	`minor_height`	`MinorHeight.lean`
Pattern as minors	`theta_eq`	`EightMain.lean`
Proposition 13.3	`plucker`, `card_cross_fiber`, `minor_vector_count`	`MinorCount.lean`
Proposition 13.4	`two_state_window_54`, `window_54`	`EightMain.lean`
Theorem 1.6	`P_le_8_27`	`EightMain.lean`

Paper	Lean	File
Lemma 14.1	`lift_row`, `lift_det`, `pivLat`	`ShortVec.lean`
Lemma 14.2	`cross_cong`, `pivLat_short`, `shortR_spec`	`ShortVec.lean`
Lemma 14.3	`basis_of_min`, `reduced_basis`, `reduced_coords`	`Reduce3.lean`
Lemma 14.4	`card_dot_dvd_le`	`ShortVec.lean`
Proposition 14.5	`card_crossSet_dot`, `short_count`	`ShortVec.lean`
Lemma 14.6	`def_class_units`, `norm_count_def`	`DefiniteNorm.lean`
Plane choice, slices	`disc3_tcomp`, `lead_eq`, `plane_one`, `plane_two`, `slice_count_tv`	`TernComp.lean`
Proposition 14.7	`ycoord_spec`, `liftM_det_ne`, `lattice_fiber_count`, `pattern_count_lattice`	`LatticeCount.lean`
Lemma 14.8	`inv_telescope`, `abel_sum`	`FortyFiveMain.lean`
Proposition 14.9	`two_state_window_152`, `window_152`	`FortyFiveMain.lean`
Theorem 1.7	`slack_le2`, `P_le_45_152`	`FortyFiveMain.lean`

Some details of the formalisation:

- The algebraic identities (Lemmas 3.2, 3.3 and (7)) are proved over $\mathbb{Z}$ by the tactics `ring` and `linear_combination`, the latter with the explicit coefficients above for Lemma 3.2. They are then transferred to $\mathbb{N}$ by writing $z = w + f$, $y = z + e$, $x = y + d$, which avoids truncated subtraction.
- The curvature is the natural number $\mathtt{bD}\, a\, x = (x - y)v - (y - z)u$. Lean first proves $eu < dv$, so this truncated subtraction is the true value.
- The counting arguments use `Finset.card_le_mul_card_image_of_maps_to`: the number of good indices is at most the size of the parameter box times the largest fiber.

- The divisor bound $\tau(n)^k \le C_k n$ (Lemma 2.2) is the file `DivisorPowerBound.lean` from the `openai/math` repository [9] (Apache-2.0), with unchanged proof bodies; imports and module visibility are adapted.
- Lemma 7.1(2) is checked by `linear_combination` with the coefficients given in its proof, which were found by polynomial division. Corollary 7.2 proves $L \ge 1$ directly from the identity, since L is an integer by definition.
- The identities $I_2 = d$, $I_3 = D$, $I_4 = L$ are checked by `linear_combination`, with certificates generated by polynomial division. The moment entries $h_i(s)$ are computed in Lean from the recursion of Lemma 10.1(3).
- Mathlib has no Hadamard inequality for determinants, so `Hadamard.lean` proves it: rescale to unit diagonal, and use $\prod \lambda_j \le \prod e^{\lambda_j - 1} = e^{\mathrm{tr} - n} = 1$ for the eigenvalues.
- The identities of Lemmas 11.1, 11.3 and 11.4 are checked by the tactics `ring` and `linear_combination`. The certificate for Lemma 11.4 has about 350 terms and was found by polynomial division. Root counting uses the Mathlib lemma `card_roots'`.
- Lemma 12.3 is proved in full in `NormCount.lean`: the prime-power root count, the Chinese remainder theorem, irrationality of $\sqrt{d}$ (`irrational_sqrt_natCast_iff`) and the separation $\varepsilon \ge 3$.
- Lemma 14.3 is proved with integer arithmetic only: the minimal determinant and the lexicographic minimum are taken with `Nat.find`, and the rounding steps use Euclidean division.
- The project has about 11700 lines of Lean. `lake build` rebuilds it in a few minutes once Mathlib is built.

16. Limits of the method

Proposition 7.3 removes the divisor loss from the local bound, but on its own the exponent is still 3/10. The local bound $T \ll a^{1/2} A^{-1/3}$ and the drop bound $T \ll A^{1/2}$ (up to the divisor factor) both allow $T \asymp a^{3/10}$ at $A \asymp a^{3/5}$. Even an ideal short-interval divisor estimate with G constant leaves this balance unchanged. So sharpening constants or improving the divisor counts cannot give a logarithmic bound. Further progress needs to restrict *which* configurations of the parameters (u, v, w, z, D, E, F) can occur together along a single orbit. Theorem 1.3 is one restriction of this kind: a long orbit must have many rises in its quotient increments. Theorem 10.2 supplies the natural candidates for further ones, the higher invariants I_r. Section 11 uses I_4 in exactly this way.

The count of Section 12 loses through the number of patterns, X^7, reduced to X^5 in Section 13 and weighted by the lattices Λ_p in Section 14. With patterns at four consecutive states one has thirteen integers and as many equations as unknown increments, so one might hope for $O(1)$ configurations per pattern. But the four-state equations have positive-dimensional families of solutions. For instance, the Newton coefficients $(\alpha, -3\alpha - 1, 3\alpha + 3, -\alpha - 3, 1)$, with coefficient polynomial $(z-1)^3(z-\alpha)$, at every state, together with constant increments $k_j = t$, satisfy all the linear and quadratic relations for every t. Such families are excluded only by the remainder equations, which the count above discards. Section 14 uses only the first minimum of Λ_p, and slices in a single direction; the other two minima and the shape of the quadric in the lattice are not used.

There is no obvious monotonicity to exploit. The curvatures D_i of successive blocks need not decrease. For $a = 13242$ the orbit of 255 begins

$$255,\ 237,\ 207,\ 201,\ 177,\ 144,\ 138,\ 132,\ \ldots$$

with quotients $51, 55, 63, 65, 74, 91, 95, 100, \ldots$. The curvatures of the first five blocks are $24, 12, 6, 111, 30$, all for blocks that lie within a factor of two.

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